REACH IS A PROPERTY OF THE HEAD: SCALE-COVARIANT ATTENTION FOR HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers trained near one megapixel can generate 4K images without retraining once their attention is adapted. Each such method must set how far every attention head looks, its reach, at a size the model never saw. Current methods share one schedule across heads or search a setting per size. We show that reach is a property of the head: at the training resolution of FLUX.1-dev, the head explains 97.5% of the variance of reach and the denoising step 2.5%. We introduce Scale-Covariant Attention (SCA), which mixes a native-size window and a native-size grid over the whole image with one number per head. Reach emulation fits that number once at 1024^2, where full attention still works, in closed form. Inside HRDiT's sampler, SCA makes a 4096^2 image in 49.2 s, $4.1\times$ faster than AReA, and beats AReA on all ten per-image metrics at three seeds. It is statistically indistinguishable from HRDiT on three preference scores in 46% of HRDiT's time. Code: github.com/shreshthsaini/scale-covariant-attention.

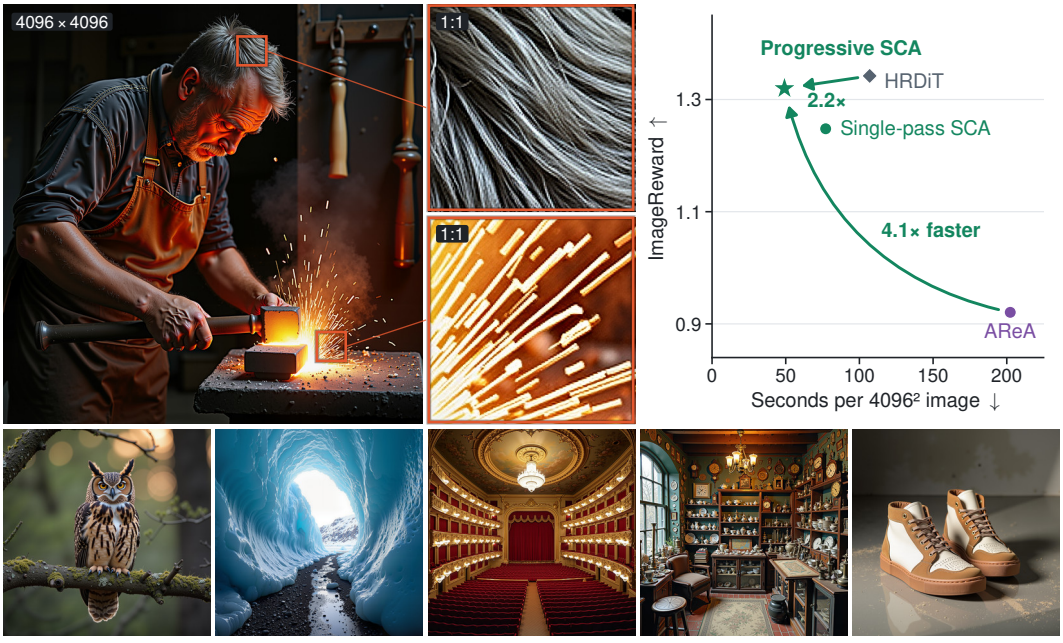


Figure 1: **Progressive SCA makes 4096^2 images with native detail, $4.1\times$ faster than AReA.** Images: FLUX.1-dev, one purpose-written prompt per category drawn with seed 0; red boxes mark native 1:1 crops at the highest edge energy. Plot: ImageReward ($n = 195$) against clean seconds per image on one GH200; arrows give clean-over-clean speedups. Figure G.1 shows more.

1 INTRODUCTION

Diffusion transformers (Peebles & Xie, 2022) such as FLUX.1 (Black Forest Labs, 2024) are trained near one megapixel. At 4096^2 an image has 16 times as many tokens, full attention costs about 256

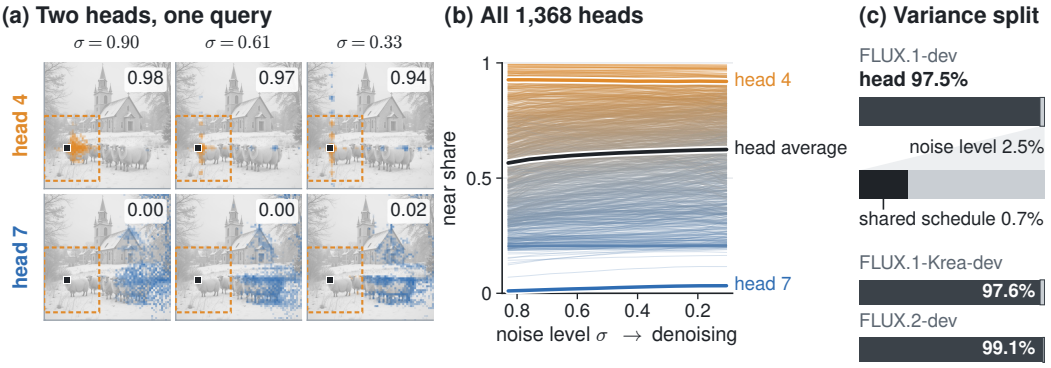


Figure 2: **Each head keeps its reach across denoising steps.** (a) Full attention of heads 4 and 7 of block 12 at one query of a 1024^2 image at three noise levels: amber within 16 tokens (dashed), blue beyond. (b) Near share of all 1,368 FLUX.1-dev heads after the first 8 steps, 32 prompts; black: head average. (c) Variance of the near share; a schedule shared by all heads can follow only the head average.

times as much, and every layer meets unseen position offsets. Full attention then returns stripes or a flat field (Figure G.2). Training-free methods therefore change the attention and must decide how far each head looks at the target size: its *reach*. AReA (Rajabi et al., 2026a) sees head-averaged attention narrow during denoising and gives all heads one schedule; HRDiT (Xue et al., 2026) searches per-head scopes for each target size; UltraImage (Zhao et al., 2025) types heads at run time. None of these choices can be checked at the target, where full attention fails. We ask whether reach can be measured where full attention works, at the training resolution, and reused at any size.

At the training resolution, reach is a property of the head (Figure 2). Over 32 prompts at 1024^2, we record each of FLUX.1-dev's 1,368 heads' share of image attention within 16 tokens of the query at every noise level after the first 8 steps. The head explains 97.5% of the variance of this share and the noise level 2.5%. The curve shared by all heads, all that one schedule for every head can follow, explains 0.7%. UltraImage noted that heads keep stable local and global roles (Zhao et al., 2025); we give the first decomposition of reach variance into heads and steps.

We introduce Scale-Covariant Attention (SCA; Figure 3). Each head mixes a native-size window around the query, for detail, and a native-size grid of keys over the whole image, for layout, with one number: its reach $\pi_h \in [0, 1]$. *Reach emulation* fits π_h at 1024^2: shrinking both views by the target's image-to-window ratio r reproduces the target's proportions, and the model's own full attention there is a valid reference. The best weight has a closed form, and one pass fills a *reach table* for every size; the table repeats on disjoint prompt halves (correlation 0.989). To our knowledge, no prior method measures per-head reach once at the training resolution and reuses it at any size.

Inside HRDiT's three-stage sampler, progressive SCA makes a 4096^2 FLUX.1-dev image in 49.2 s on one GH200, $4.1\times$ faster than AReA and $2.2\times$ faster than HRDiT (Figure 1). It beats AReA on all ten per-image metrics at each of three seeds and on HRGen-1K, with ImageReward rising from 0.92 to 1.32 ($+0.40$ $[0.32, 0.49]$, $n = 195$, seed 0). Against HRDiT, its ImageReward, HPSv2.1, and PickScore are statistically indistinguishable (every 95% interval includes zero) in 46% of HRDiT's time. A single-pass mode is $2.63\times$ faster than AReA and keeps the most crop-level detail; the progressive schedule costs patch-level fidelity, where HRDiT and single-pass SCA lead.

- We show that reach is a property of the head: at the training resolution, the head explains 97.5% to 99.1% of its variance in three FLUX models (Section 2).
- We introduce SCA, which mixes a native-size window and grid with one weight per head, and reach emulation, which fits that weight once at 1024^2 in closed form (Section 3).
- We show that progressive SCA makes a 4096^2 image in 49.2 s, $4.1\times$ faster than AReA and better on all ten per-image metrics at three seeds, and that per-head weights carry the gain (Section 4).

2 Reach is a property of the head

A head that attends near the query at one denoising step does so at the others, and a head that looks across the image keeps looking across it.

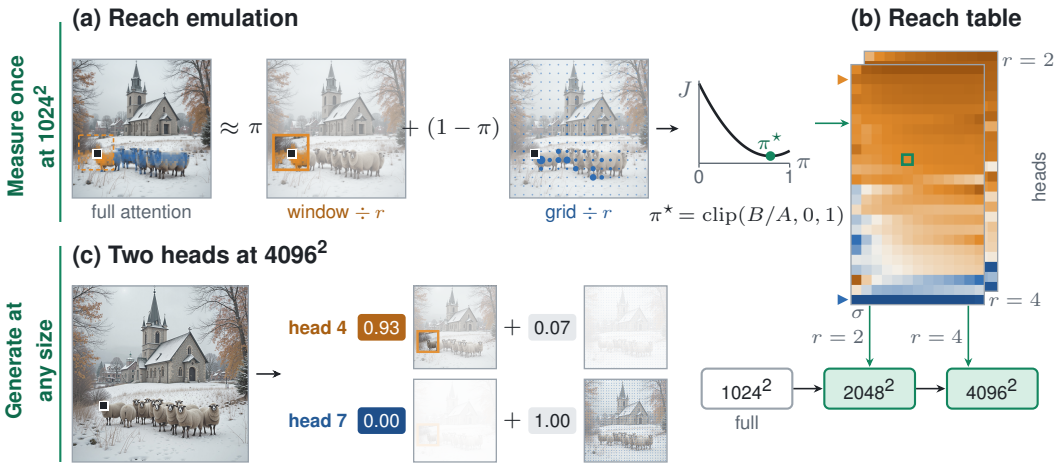


Figure 3: **Reach is measured once at 1024^2 and read at any size.** (a) Full attention of one head at 1024^2 is fitted by a mix of its window and grid views shrunk by r; the fit has a closed form. (b) The weight fills one cell of the reach table: heads by noise level, one sheet per r. (c) At 4096^2, head 4 of Figure 2 mixes mostly its window and head 7 only its grid; each sampler stage reads the table at its r. Figure C.1 zooms in.

Reach. FLUX.1 applies joint attention to text and image tokens in 57 blocks of 24 heads and is trained near 1024^2 pixels, a grid of 64×64 image tokens. For one head and a query i, split the keys into a near set around the query and a far set. The output splits exactly as $O_i^{\text{full}} = m_i\, O_i^{\text{near}} + (1 - m_i)\, O_i^{\text{far}}$, where O_i^{near} and O_i^{far} attend only to their own set and m_i is the near set's softmax mass (Lemma 1). This near share is the head's reach at that query: near 1 for a head that looks near, near 0 for one that looks far (Figure 2a). We record it within 16 tokens for 256 seeded queries per block and step, over 32 purpose-written prompts at 1024^2 outside the evaluation sets.

The head sets the reach. After the first 8 steps (noise level $\sigma < 0.89$), the head explains 97.5% of the variance of this share over all heads and noise levels. The noise level within a head explains 2.5% (Figure 2b,c). The head-averaged curve rises from 0.57 to 0.62, the narrowing AReA reports, yet explains only 0.7%; no schedule shared by all heads can explain more. FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) and FLUX.2-dev (Black Forest Labs, 2025) give 97.6% and 99.1% for the head (Figure 2c). One number per head thus captures nearly all of the variation and, because it barely moves with the step, can be measured once.

3 Scale-Covariant Attention

3.1 Two native-size views

On its native grid the model saw position offsets of at most 63 tokens per axis and 64^2 image keys per query. SCA keeps both at any size with two views: a near view O_i^{loc}, a native-size window around the query, and a far view O_i^{glob}, a native-size grid over the whole image. An image query i of head h mixes them as $O_i = \pi_h O_i^{\text{loc}} + (1-\pi_h) O_i^{\text{glob}}$, with the head's reach $\pi_h \in [0, 1]$ (Figure 3c).

Setting. A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, where $[n] = \{0, \ldots, n-1\}$. At 1024^2 the grid is $n_0 \times n_0$ with $n_0 = 64$. Rotary embeddings (Su et al., 2021) give token j a position $p_j \in \mathbb{Z}^2$ (text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at u, and a set $\mathcal{K}$ of (key, value, key position) triples, attention is

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u') \in \mathcal{K}} \operatorname{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v, \tag{1}$$

where d is the head dimension. Full attention O_i^{full} attends to all keys at their own positions; at a target grid $H = s_H n_0$, $W = s_W n_0$ (s when equal) it meets offsets up to s times the trained range.

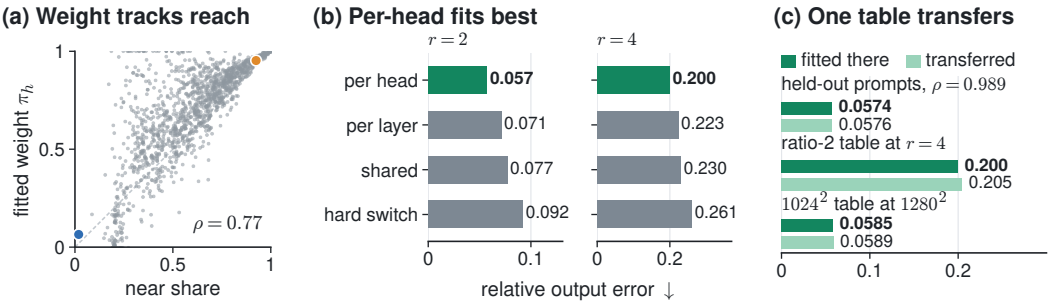


Figure 4: **The fitted weight measures reach, fits full attention best, and transfers.** (a) π_h against the near share of 1,368 heads at $\sigma = 0.62$; amber and blue: heads 4 and 7. (b) Relative output error of four mixing rules, 32 prompts. (c) A table used where it was not fitted (light) against the table fitted there (dark); held-out: each prompt half's table on the other half. Appendix B expands each panel.

Near view. Image queries form tiles of $t \times t$ tokens ($t = 8$), and the window $\mathcal{W}_w(i)$ of query i is the $w \times w$ block centred on its tile and clamped to the grid (Appendix C). Its keys are the text tokens, a bridge set $\mathcal{B}$ of image tokens, and the window, at their original positions:

$$\mathcal{K}_i^{\mathrm{loc}}(w) = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{j\in\mathcal{B}\cup\mathcal{W}_w(i)}, \qquad O_i^{\mathrm{loc}}(w) = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\mathrm{loc}}(w)\big), \quad (2)$$

where $\mathcal{K}_i^{\mathrm{loc}}(w)$ is the local key set; one FlexAttention call computes it (Dong et al., 2024). The bridge lets each window see the whole image. In each cell c of an 8×8 partition, it keeps the 16 tokens whose pre-rotation keys lie farthest from the mean image key (AReA's rule; Rajabi et al., 2026a):

$$\mathcal{B} = \bigcup_c \mathrm{top}_{16}\big\{\textstyle\sum_h \|k_j^h - \bar{k}^h\|^2 : j \in c\big\}, \quad (3)$$

where $\bar{k}^h$ is the mean image key of head h. With $w = n_0$ the window is an exact native-size sliding window, for content of fixed token size, such as edges and texture.

Far view. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\, s_W b_2 + \lfloor s_W/2 \rfloor)$, and $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ maps a position to its block:

$$\mathcal{K}^{\mathrm{glob}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b\in\mathcal{G}_0}, \qquad O_i^{\mathrm{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\mathrm{glob}}\big), \quad (4)$$

where text queries sit at $c(p_i) = 0$. This is AReA's first phase at pooling factor s, for content that spans a fixed fraction of the image, such as layout. Every offset of the grid and of the native window stays in the trained range, and the cost is linear in N (Propositions 2 and 3). At 4096^2, SCA scores 6.9 times fewer query-key pairs than full attention (full version).

3.2 Reach Emulation

Full attention fails at the target, so nothing there can score π_h; reach emulation scores it on a miniature of the target at 1024^2, against the model's own full attention (Figure 3a). On the native $n \times n$ grid ($n = 64$), emulated ratio r uses subsampling factor r, window side n/r, and tile side $\mathrm{round}(tn/(rn_0))$, so the image-to-window ratio is r, as at a target r times larger. The sampler runs unmodified full attention (Algorithm 1).

Closed-form fit. For block l, head h, noise bin β, and ratio r, we accumulate three sums over prompts, steps with $\sigma \in \beta$, image queries, and channels:

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}},\, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \quad (5)$$

where O^{loc} and O^{glob} are the emulated views without the bridge. A weight π then has squared error $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 1 (Closed-form fit and transfer cost)**.** *In each cell (l, h, β, r), J is minimized on $[0, 1]$ by $\pi^\star = \mathrm{clip}(B/A, 0, 1)$, with $\pi^\star = 1$ if $A = 0$. Any other weight $\pi' \in [0, 1]$ constant in the cell has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$. This covers weights fitted at another ratio or resolution, per layer, or shared, and a hard switch.*

One pass collects ratios 2 and 4 and fills the reach table $\pi_{lh}(\sigma; r)$, and its sums score every other rule without another pass. We assume that the weight depends only on r, which is scale covariance.

Algorithm 1: **Reach emulation**, once per checkpoint

```
# Input: emulation prompts, model at 1024^2
# Output: reach table π[r, l, h, β]; App. B
def emulate(q, k, v, l, sigma):  # attn hook
  O_full = attn(q, k, v)         # Eq. 1, all keys
  beta = noise_bin(sigma)        # 14 bins, App. B
  for r in (2, 4):               # r times larger
    O_loc = window(q, k, v, n // r)  # Eq. 2
    O_glob = grid(q, k, v, stride=r) # Eq. 4
    D, F = O_loc - O_glob, O_full - O_glob
    A[r, l, :, beta] += sum(D * D)   # Eq. 5
    B[r, l, :, beta] += sum(F * D)   # per head
  return O_full                  # sampler as is
pi = clip(B / A, 0, 1)           # Prop. 1
```

Algorithm 2: **SCA attention**, one call at any size

```
# Input: q, k, v (T text, H x W image), l, σ, π
# Output: all T + HW outputs; App. C
Kg = text + centres(k, v, H // n0)  # Eq. 4
O_glob = attn(q, Kg, at=c(p))  # coarse grid
if sigma > sigma_g:            # opening steps
  return O_glob                # global only
w = window_side(sigma)         # 128 to 80, n0
Kl = text + bridge(k) + window(w)  # Eq. 2, 3
O_loc = flex_attention(q, Kl)  # App. C
r = max(H, W) / w              # window ratio
pi = interp(table[l], sigma, r)  # Eq. 7
O = pi * O_loc + (1 - pi) * O_glob[img]  # Eq. 6
return [O_glob[txt], O]        # text: global
```

Table 1: **Main comparison at 4096^2.** FLUX.1-dev, 195 Aesthetic-4K prompts, seed 0; HPSv2.1 $\times 100$; M_c: MUSIQ on 1024^2 crops; FID_p: patch FID†. Time: clean batch-1 median on one GH200 (‡campaign mean) and its ratio to progressive SCA's of the same kind. $+/-$: per-image metrics, of ten, on which progressive SCA is significantly better / worse than the row after Holm's correction. Bold: best, underline: second per group; shaded: ours; Table E.1 has every metric. Key result: progressive SCA is the fastest method, and no baseline is significantly better on any per-image metric.

	Time per image		Preference ↑			Detail ↑	Set level ↓		Ten metrics
Method	s ↓	vs. ours	IR	HPSv2.1	Pick	M_c	FID	$FID_p^\dagger$	$+/-$
Single pass, AReA's sampler (28 steps)									
Full attention	186.5	3.8×	−2.18	7.60	17.94	33.1	312.4	252.2	10 / 0
DyPE	195.9‡	3.9×‡	−0.10	20.55	20.51	50.3	205.5	171.9	10 / 0
SEGA	190.4‡	3.8×‡	0.56	24.34	21.64	50.3	164.9	125.8	10 / 0
AReA	202.4	4.1×	0.92	27.16	22.36	50.4	145.8	102.9	10 / 0
Single-pass SCA (ours)	**77.1**	1.6×	**1.25**	**29.73**	**23.12**	**60.1**	**140.2**	**98.4**	7 / 2
Multi-stage, own recipes (UltraImage 30+30 steps; HRDiT and ours 30+17+10)									
UltraImage	308.5‡	6.1×‡	1.28	30.77	23.33	41.7	142.0	109.5	9 / 0
HRDiT	107.0	2.2×	**1.34**	30.92	**23.46**	53.5	**138.4**	**103.0**	2 / 0
Progressive SCA (ours)	**49.2**	1.0×	1.32	**31.02**	23.41	**54.1**	141.9	107.7	*ref.*

One table serves every size. Tables fitted on disjoint prompt halves correlate at 0.989 at ratio 2 and 0.986 at ratio 4, and the fitted weight tracks each head's measured near share ($\rho = 0.77$; Figure 4). In relative output error, $\sum J / \sum \|O^{\text{full}}\|^2$, the ratio-2 table applied at ratio 4 scores 0.205 against 0.200 for the ratio-4 table, below every coarser rule fitted at ratio 4. The 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585, although the two resolutions emulate ratio 2 with windows of 32 and 40 tokens. The weights depend on the ratio, as scale covariance assumes.

3.3 Sampling with the reach table

Progressive SCA. A sampler reads the table at every size it visits (Algorithm 2). Our main configuration keeps HRDiT's three-stage sampler (Xue et al., 2026) and replaces only its four attention changes (NTK-scaled RoPE, a proportional attention scale, per-head key pruning, and position alignment) with SCA (Figure 3c). HRDiT denoises 30 steps at 1024^2 with the model's own attention, then re-noises the upsampled image and runs the last 17 steps of its schedule at 2048^2 and the last 10 at 4096^2. These stages read the table at ratios 2 and 4, from $\sigma = 0.81$ and 0.61 on.

Single-pass mode. SCA also samples at the target in one pass of 28 steps. A window schedule $w(\sigma)$ sets the ratio $r(\sigma) = \max(H, W)/w(\sigma)$, and an image query takes

$$O_i \;=\; \pi\, O_i^{\text{loc}}\big(w(\sigma)\big) + (1-\pi)\, O_i^{\text{glob}} \;\text{ if } \sigma \le \sigma_g, \qquad O_i = O_i^{\text{glob}} \;\text{ if } \sigma > \sigma_g, \tag{6}$$

where $\pi = \pi_{lh}(\sigma; r(\sigma))$, $\sigma_g = 0.89$, and text queries take O_i^{glob}. Between ratios 2 and 4, π is linear in $\log r$, and in σ it is interpolated between occupied noise bins:

$$\pi_{lh}(\sigma; r) = (1-\lambda)\,\pi_{lh}(\sigma; 2) + \lambda\,\pi_{lh}(\sigma; 4), \qquad \lambda = \text{clip}(\log_2 r - 1,\, 0,\, 1), \tag{7}$$

where λ is the weight of the ratio-4 table. The global view runs alone for the first 8 of 28 steps, as in AReA, because output error cannot see layout (Section 4.3). Windows of 128, 112, 96, and 80

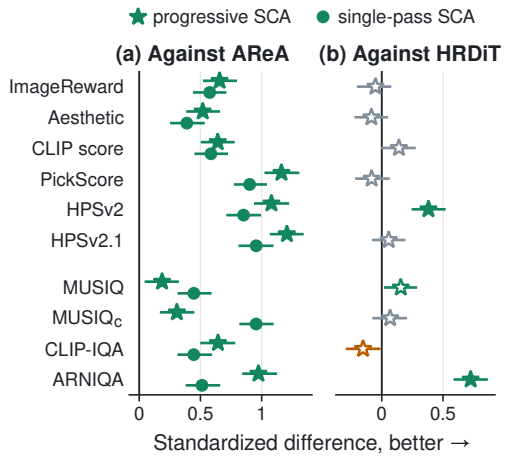


Figure 5: **Both modes beat AReA on all ten per-image metrics; progressive SCA is worse than HRDiT on none.** Paired differences ($n = 195$) over the SD of the per-prompt differences, 95% intervals: (a) both modes minus AReA, (b) progressive SCA minus HRDiT. Filled: significant after Holm's correction; open: before it only (Table E.2).

Table 2: **Generality.** Paired differences at 4096^2 (HPS: HPSv2.1×100), coloured as in Section 4. Holm +/−: per-image metrics, of ten, on which the first-named method is significantly better / worse. The full version gives every metric with intervals. Key result: no setting is significantly worse on any metric.

		Paired difference ↑			Holm
Setting	n	IR	HPS	Pick	+/−
Progressive SCA − AReA					
Seed 0	195	+0.40	+3.86	+1.05	10/0
Seed 1	195	+0.37	+3.67	+1.00	10/0
Seed 2	195	+0.35	+3.61	+0.99	10/0
HRGen-1K	1000	+0.90	+6.59	+1.40	10/0
Single-pass SCA − AReA					
Seed 0	195	+0.33	+2.56	+0.76	10/0
Seed 1	195	+0.31	+2.49	+0.77	10/0
Seed 2	195	+0.26	+2.53	+0.74	10/0
HRGen-1K	1000	+0.76	+4.60	+1.09	10/0
FLUX.1-Krea	195	+0.02	+0.91	+0.17	7/0
Without the schedule − full attention					
FLUX.2-dev	50	+0.46	+1.84	+0.56	9/0

tokens then take one step each before the native $w = n_0$. The gate, the window sizes, and the 1,024 bridge tokens are set by hand; only the weights are measured (full version). With $w = n_0$ right after the gate and no bridge, the operator is SCA without the schedule.

4 Experiments

Data. We follow the field's protocol at 4096^2: the 195-prompt Aesthetic-4K split with its 4K references (Zhang et al., 2025a), as in AReA, DyPE, TIDE, and SigMa (Rajabi et al., 2026a; Issachar et al., 2025; Liu et al., 2026; Zhao et al., 2026), and the 1000 HRGen-1K prompts. Seed labels 0, 1, and 2 are shared by all methods. Four prompts (boats, Moai, woman and bear, underwater surfer) were used in development, which favours SCA slightly; a fixed 46-prompt subset chosen before any ablation serves the ablations.

Models and baselines. The main model is FLUX.1-dev, and every baseline runs from its released code (Table 1). Single-pass SCA, full attention, AReA, DyPE, SEGA (Rajabi et al., 2026b), and SigMa share AReA's sampler, 28 Euler steps at guidance 4.5 (Appendix D); our AReA run reproduces its published FLUX.1-Krea-dev scores within 0.3 (full version). Progressive SCA, UltraImage, and HRDiT use multi-stage recipes (Appendix D). One reach table per checkpoint (32 prompts for FLUX.1-dev, 16 for the others) serves every size and both modes.

Metrics and timing. The ten per-image metrics are ImageReward, HPSv2, HPSv2.1, PickScore, CLIP score, aesthetic score (Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023; Radford et al., 2021; Schuhmann, 2022), MUSIQ on the full image and over four 1024^2 crops (MUSIQ$_c$), CLIP-IQA, and ARNIQA (Ke et al., 2021; Wang et al., 2022; Agnolucci et al., 2024). FID and KID use all 195 references at 299^2, as in AReA (Heusel et al., 2017); patch FID and KID use the same crops, and patch FID is our own definition. Intervals are 95% paired percentile bootstraps over prompts (Appendix D). A method beats another on a metric when that interval excludes zero, which tables mark blue in the row's favour and orange against it. Holm's correction over the ten per-image metrics leaves every "all ten" claim intact (Tables 1 and 2). Clean latency is the batch-1 median of three prompts on one idle GH200 after a warm-up image, a point estimate, and a speed ratio divides two times of the same kind (Appendix D).

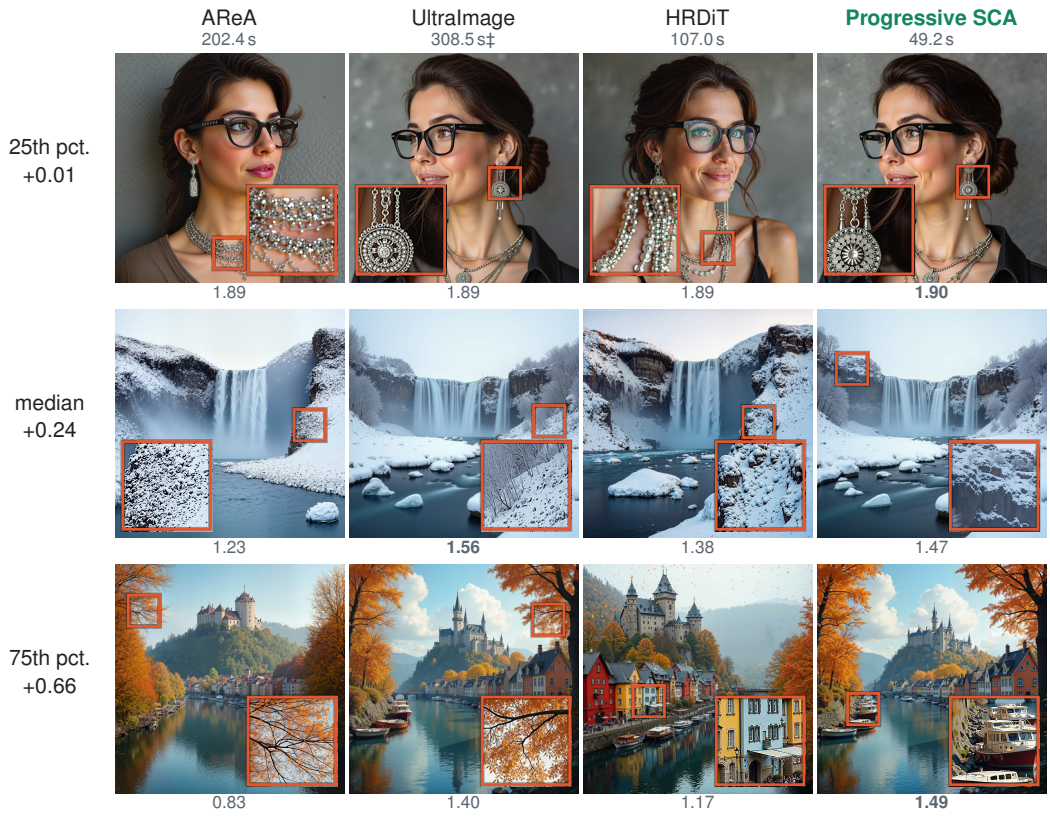


Figure 6: **Progressive SCA's gain over AReA is in the whole image, in a quarter of AReA's time, and it is level with HRDiT.** Rows: the evaluation prompts at the 25th, 50th, and 75th percentile of the paired ImageReward difference, progressive SCA minus AReA (seed 0, $n = 195$, mean $+0.40$). Numbers: each image's ImageReward, best bold; insets: each image's 512×512 window of highest edge energy at native pixels; heads: seconds per image (‡campaign mean). Over all prompts, ImageReward differs from HRDiT's by -0.02 $[-0.09, 0.03]$. Figure G.1 shows more.

4.1 Main results

Progressive SCA beats AReA on all ten per-image metrics at $4.1\times$ its speed. ImageReward rises from 0.92 to 1.32 ($+0.40$ $[0.32, 0.49]$), PickScore by $+1.05$ $[0.93, 1.19]$, and HPSv2.1 by $+3.86$ $[3.41, 4.30]$ ($n = 195$; Table 1, Figure 5). Seeds 1 and 2 repeat the result on all ten (ImageReward $+0.37$ $[0.29, 0.46]$ and $+0.35$ $[0.27, 0.45]$), and so does HRGen-1K ($+0.90$ $[0.84, 0.96]$, $n = 1000$; Table 2). Progressive SCA also beats DyPE, SEGA, and full attention on all ten and UltraImage on nine (Table 1). Figure 6 shows that the gain lies in the whole image, on the prompts at the quartiles of the ImageReward difference.

Against HRDiT, progressive SCA is statistically indistinguishable on three preference scores in 46% of its time. Both follow HRDiT's schedule, guidance, and seeds and differ in attention, precision (fp16 for HRDiT's runner, bf16 for ours), and runner. The differences are -0.02 $[-0.09, 0.03]$ in ImageReward, $+0.10$ $[-0.14, 0.35]$ in HPSv2.1, and -0.05 $[-0.14, 0.04]$ in PickScore ($n = 195$); intervals that include zero do not establish equivalence. After Holm's correction, progressive SCA remains better on HPSv2 ($+0.38$ $[0.24, 0.51]$) and ARNIQA ($+2.9$ $[2.3, 3.4]$) and is worse on none. MUSIQ and CLIP-IQA differ only before the correction (Table E.2). Crop detail is statistically indistinguishable from HRDiT's (MUSIQ_c $+0.6$ $[-0.7, 1.8]$; Figure E.1).

Single-pass SCA is $2.63\times$ faster than AReA and keeps the most crop-level detail. It beats AReA on 12 of 15 metrics, all ten per-image ones among them, with ImageReward $+0.33$ $[0.25, 0.41]$ and FID from 145.8 to 140.2 (-5.6 $[-10.6, -0.4]$). KID, patch FID, and IS are unresolved (full version). DyPE and SEGA trail it on all ten per-image metrics, and SigMa on every one it was scored on.

Table 3: **Ablations at 4096^2.** Each group opens with its shaded per-head reference (absolute values), and rows below give the variant minus it (views: minus the first shaded row), coloured as in Section 4. 46-prompt FID_p values are point estimates. $+/-$: per-image metrics, of ten, on which per-head weights are significantly better / worse after Holm's correction. The full version gives every metric. Key result: per-head weights beat coarser ones on per-image metrics in both samplers but have higher patch FID inside the progressive one.

	Preference ↑			Detail ↑	Set level ↓	Ten metrics
Variant	IR	HPSv2.1	Pick	M_c	FID_p	$+/-$
Reach weights, single pass without the schedule (46 prompts)						
per-head π_h (ours)	1.12	29.98	22.96	62.5	168.0	*ref.*
shared $\pi(\sigma)$	−0.09	−1.83	−0.38	−10.4	−1.3	9 / 0
per-layer $\pi_\ell(\sigma)$	−0.05	−0.70	−0.16	−2.4	−0.3	5 / 0
Reach weights, progressive SCA (195 prompts)						
per-head π_h (ours)	1.32	31.02	23.41	54.1	107.7	*ref.*
shared $\pi(\sigma)$	−0.01	−0.38	−0.07	−4.3	−7.3	10 / 0
per-layer $\pi_\ell(\sigma)$	−0.01	−0.39	−0.07	−2.8	−6.4	10 / 0
window only ($\pi = 1$)	−0.02	−0.36	−0.14	+0.5	+6.0	6 / 1
Views, single pass without the schedule (46 prompts; against the first shaded row)						
no opening	−1.33	−4.07	−2.18	+2.5	+34.6	6 / 2
local only	−0.36	−2.86	−1.20	−0.3	+24.2	7 / 0
global only	−0.40	−5.66	−1.44	−11.7	+69.0	10 / 0
mean pooling	−1.52	−9.75	−2.60	−4.2	+16.6	8 / 0

The progressive schedule trades patch-level fidelity for preference. Against single-pass SCA, it gains ImageReward ($+0.07$ $[0.01, 0.13]$) but loses MUSIQ_c (-6.0 $[-7.3, -4.6]$) and patch FID ($+9.2$ $[5.1, 13.2]$). Its patch FID also exceeds HRDiT's ($+4.6$ $[0.6, 7.7]$) and, at two of three seeds, than AReA's (full version). Its MUSIQ_c still beats AReA's ($+3.7$ $[2.0, 5.4]$).

4.2 Generality, other sizes, and latency

Time grows linearly with pixel count. Single-pass SCA is faster than every baseline we timed at every size (Figure 7). Fitting clean seconds $\propto N^b$ gives $b = 0.99$ against 1.12 for AReA, and at 8192^2 SCA takes 309.7 s against AReA's 952.0 s. Quality depends on the aspect ratio: at 2048×6144 single-pass SCA beats AReA on seven of 15 metrics, but at 2048×4096 it wins three and loses three (full version).

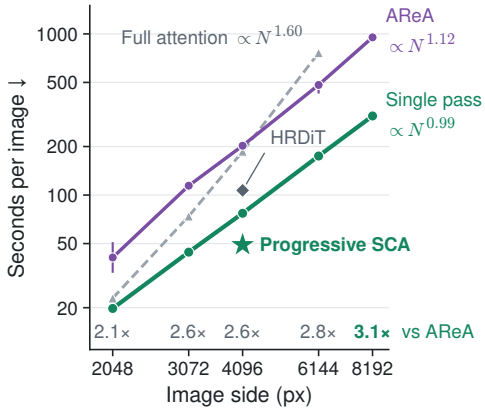


Figure 7: **Single-pass SCA's time grows linearly with pixel count.** Clean batch-1 seconds per image on one GH200; labels give b of seconds $\propto N^b$, the bottom row AReA's time over single-pass SCA's. Figure E.3 adds the speed-up over full attention.

Gains hold across seeds, prompt sets, and checkpoints. Single-pass SCA's gains over AReA hold at seeds 1 and 2 and on HRGen-1K (Table 2). Reach tables fitted on 4 to 16 prompts move ImageReward by at most 0.012 without the schedule. On FLUX.1-Krea-dev, AReA's own checkpoint, single-pass SCA wins seven of ten per-image metrics and loses none. Its ImageReward is unresolved ($+0.020$ $[-0.020, 0.063]$) and its patch FID is worse ($+3.4$ $[1.6, 7.1]$; full version). On FLUX.2-dev, SCA without the schedule beats full attention on nine of ten.

4.3 The measured reach carries the gain

Per-head weights beat coarser ones in images (Table 3). At 4096^2, one shared weight per noise level changes ImageReward by -0.09 $[-0.17, -0.01]$ and MUSIQ_c by -10.4, and per-layer weights still lose five of ten per-image metrics. Inside the progressive sampler, per-head weights still win all ten per-image metrics against both after Holm's correction, by smaller margins, but have higher patch FID.

Output error shows the same order. In emulation at ratio 4, the per-head mix has relative error 0.200, per-layer weights 0.223, one weight per noise level 0.230, and the best hard switch 0.261. Ratio 2 keeps the order (Figure 4b). By Proposition 1, a coarser rule pays at least $\sum A(\pi' - \pi^\star)^2$, which grows with the spread of reach among merged heads.

Both views and the opening are needed. Local-only attention in the last 20 steps changes ImageReward by -0.36 $[-0.52, -0.21]$, global-only attention loses all ten per-image metrics, and mean pooling in place of block centres changes ImageReward by -1.52. Output error cannot see layout. Fitted on it alone, the best hard switch is local attention from the first step, yet dropping the opening changes ImageReward by -1.33 $[-1.60, -1.06]$. The global opening, or the 1024^2 stage of progressive SCA, therefore sets the layout. Without the hand-set schedule and bridge, MUSIQ_c changes by $+1.7$ and no other column moves significantly (full version).

5 Related work

Training-free high-resolution sampling. Methods alter convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024) or split them by frequency (Qiu et al., 2024), add native-resolution guidance (Du et al., 2024; Bu et al., 2025; Koh et al., 2025), or adapt RoPE (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2026; Rajabi et al., 2026b; Liu et al., 2026). AReA (Rajabi et al., 2026a) sets one schedule for all heads by hand, in our notation a window weight of 0 and then 1 for every head; SCA borrows its opening, block-centre keys, and bridge selection. ResDiT also mixes position-scaled global and native-size patch attention (Ma et al., 2025).

Head-specific attention. HRDiT searches per-head scopes under a cost budget for each target size (Xue et al., 2026), and UltraImage types heads by entropy at run time and noted stable local and global head roles (Zhao et al., 2025). Sparse attention varies windows, spans, or key scales per layer or head in diffusion transformers (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b), language models (Fu et al., 2024; Xiao et al., 2024), and vision transformers (Pan et al., 2022). These settings are fitted against dense attention at the size where they run, which at 4096^2 is the failure. Head roles, two branches (our local branch is neighbourhood attention; Hassani et al., 2022), a global opening, and shrinking windows have precedents.

6 Limitations and conclusion

Limitations. Progressive SCA's patch FID is higher than HRDiT's and single-pass SCA's, and inside its sampler coarser weights have lower patch FID than per-head weights. In single-pass mode, small additional figures occasionally appear in scenes with people or animals; the progressive configuration reduces this to the level of the multi-stage baselines. The opening, window sizes, and bridge are set by hand, and the window sizes were chosen on the 46-prompt subset, part of the evaluation set. Progressive SCA depends on HRDiT's sampler, which does not support FLUX.1-Krea-dev, so the 6144^2, 8192^2, FLUX.2-dev, and sensitivity results use SCA without the schedule. Reach emulation omits the bridge and sees 16 times fewer keys per view than the target, so the deployed operator's output error is untested. All metrics are automatic.

Conclusion. Reach is a property of the head, so SCA fits it once per head at the training resolution, where full attention is still correct, and reuses it at any size. Inside HRDiT's sampler it is $4.1\times$ faster than AReA at 4096^2, better on all ten per-image metrics at three seeds, and statistically indistinguishable from HRDiT on three preference scores in 46% of HRDiT's time. Code: github.com/shreshthsaini/scale-covariant-attention.

REPRODUCIBILITY STATEMENT

Section 4 states the protocol, and Appendix D summarizes how each baseline was run and how intervals and times are computed. Appendix B gives the noise bins and the analysis of reach emulation, and Algorithms 1 and 2 give the procedures. The full version of this paper (62 pages) adds the complete pseudocode with progressive SCA, every hyperparameter, the kernel layout and numerical checks, each baseline's code revision and recipe, the logged compute, the emulation prompts, and a table that maps every table and figure to the script and results file that produce it. Code, reach tables, per-image metrics, and paired-difference files will be released with the code (link in the abstract).

REFERENCES

Lorenzo Agnolucci, Leonardo Galteri, Marco Bertini, and Alberto Del Bimbo. ARNIQA: Learning distortion manifold for image quality assessment. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

Black Forest Labs. FLUX.1. `https://github.com/black-forest-labs/flux`, 2024.

Black Forest Labs. FLUX.2. `https://huggingface.co/black-forest-labs/FLUX.2-dev`, 2025.

Black Forest Labs and Krea AI. FLUX.1 Krea [dev]. `https://huggingface.co/black-forest-labs/FLUX.1-Krea-dev`, 2025.

Jiazi Bu, Pengyang Ling, Yujie Zhou, Pan Zhang, Tong Wu, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, et al. HiFlow: Training-free high-resolution image generation with flow-aligned guidance. *arXiv preprint arXiv:2504.06232*, 2025.

Juechu Dong, Boyuan Feng, Driss Guessous, Yanbo Liang, and Horace He. Flex attention: A programming model for generating optimized attention kernels. *arXiv preprint arXiv:2412.05496*, 2024.

Ruoyi Du, Dongliang Chang, Timothy Hospedales, Yi-Zhe Song, and Zhanyu Ma. DemoFusion: Democratising high-resolution image generation with no $$$. *arXiv preprint arXiv:2311.16973*, 2023.

Ruoyi Du, Dongyang Liu, Le Zhuo, Qin Qi, Hongsheng Li, Zhanyu Ma, and Peng Gao. I-Max: Maximize the resolution potential of pre-trained rectified flow transformers with projected flow. *arXiv preprint arXiv:2410.07536*, 2024.

Tianyu Fu, Haofeng Huang, Xuefei Ning, Genghan Zhang, Boju Chen, Tianqi Wu, Hongyi Wang, Zixiao Huang, et al. Mixture of attention spans: Optimizing LLM inference efficiency with heterogeneous sliding-window lengths. *arXiv preprint arXiv:2406.14909*, 2024.

Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. Neighborhood attention transformer. *arXiv preprint arXiv:2204.07143*, 2022.

Yingqing He, Shaoshu Yang, Haoxin Chen, Xiaodong Cun, Menghan Xia, Yong Zhang, Xintao Wang, Ran He, et al. ScaleCrafter: Tuning-free higher-resolution visual generation with diffusion models. *arXiv preprint arXiv:2310.07702*, 2023.

Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *arXiv preprint arXiv:1706.08500*, 2017.

Linjiang Huang, Rongyao Fang, Aiping Zhang, Guanglu Song, Si Liu, Yu Liu, and Hongsheng Li. FouriScale: A frequency perspective on training-free high-resolution image synthesis. *arXiv preprint arXiv:2403.12963*, 2024.

Noam Issachar, Guy Yariv, Sagie Benaim, Yossi Adi, Dani Lischinski, and Raanan Fattal. DyPE: Dynamic position extrapolation for ultra high resolution diffusion. *arXiv preprint arXiv:2510.20766*, 2025.

Zhiyu Jin, Xuli Shen, Bin Li, and Xiangyang Xue. Training-free diffusion model adaptation for variable-sized text-to-image synthesis. *arXiv preprint arXiv:2306.08645*, 2023.

Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. MUSIQ: Multi-scale image quality transformer. *arXiv preprint arXiv:2108.05997*, 2021.

Yuval Kirstain, Adam Polyak, Uriel Singer, Shahbuland Matiana, Joe Penna, and Omer Levy. Pick-a-Pic: An open dataset of user preferences for text-to-image generation. *arXiv preprint arXiv:2305.01569*, 2023.

Sungho Koh, SeungJu Cha, Hyunwoo Oh, Kwanyoung Lee, and Dong-Jin Kim. ScaleDiff: Higher-resolution image synthesis via efficient and model-agnostic diffusion. *arXiv preprint arXiv:2510.25818*, 2025.

Yihua Liu, Fanjiang Ye, Bowen Lin, Rongyu Fang, and Chengming Zhang. TIDE: Text-informed dynamic extrapolation with step-aware temperature control for diffusion transformers. *arXiv preprint arXiv:2603.08928*, 2026.

Yiyang Ma, Feng Zhou, Xuedan Yin, Pu Cao, Yonghao Dang, and Jianqin Yin. ResDiT: Evoking the intrinsic resolution scalability in diffusion transformers. *arXiv preprint arXiv:2512.01426*, 2025. URL https://arxiv.org/abs/2512.01426.

Zizheng Pan, Jianfei Cai, and Bohan Zhuang. Fast vision transformers with HiLo attention. *arXiv preprint arXiv:2205.13213*, 2022.

William Peebles and Saining Xie. Scalable diffusion models with transformers. *arXiv preprint arXiv:2212.09748*, 2022.

Haonan Qiu, Shiwei Zhang, Yujie Wei, Ruihang Chu, Hangjie Yuan, Xiang Wang, Yingya Zhang, and Ziwei Liu. FreeScale: Unleashing the resolution of diffusion models via tuning-free scale fusion. *arXiv preprint arXiv:2412.09626*, 2024.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, et al. Learning transferable visual models from natural language supervision. *arXiv preprint arXiv:2103.00020*, 2021.

Javad Rajabi, Koorosh Roohi, David B. Lindell, and Babak Taati. AReA: Adaptive receptive-field attention for efficient high-resolution diffusion transformers. *arXiv preprint arXiv:2610.17227*, 2026a. URL https://arxiv.org/abs/2610.17227.

Javad Rajabi, Kimia Shaban, Koorosh Roohi, David B. Lindell, and Babak Taati. SEGA: Spectral-energy guided attention for resolution extrapolation in diffusion transformers. *arXiv preprint arXiv:2605.22668*, 2026b.

Christoph Schuhmann. LAION aesthetics predictor. https://github.com/LAION-AI/aesthetic-predictor, 2022.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021.

Jianyi Wang, Kelvin C. K. Chan, and Chen Change Loy. Exploring CLIP for assessing the look and feel of images. *arXiv preprint arXiv:2207.12396*, 2022.

Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341*, 2023.

Haocheng Xi, Shuo Yang, Yilong Zhao, Chenfeng Xu, Muyang Li, Xiuyu Li, Yujun Lin, Han Cai, et al. Sparse VideoGen: Accelerating video diffusion transformers with spatial-temporal sparsity. *arXiv preprint arXiv:2502.01776*, 2025.

Guangxuan Xiao, Jiaming Tang, Jingwei Zuo, Junxian Guo, Shang Yang, Haotian Tang, Yao Fu, and Song Han. DuoAttention: Efficient long-context LLM inference with retrieval and streaming heads. *arXiv preprint arXiv:2410.10819*, 2024.

Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. ImageReward: Learning and evaluating human preferences for text-to-image generation. *arXiv preprint arXiv:2304.05977*, 2023.

Yu Xue, Haoxuan Qu, Zhuoling Li, Hongbin Xu, Jianxiong Yin, Simon See, Hossein Rahmani, and Jun Liu. HRDiT: Training-free high-resolution image generation with off-the-shelf diffusion transformer models. *arXiv preprint arXiv:2608.07003*, 2026. URL https://arxiv.org/abs/2608.07003.

Haosen Yang, Adrian Bulat, Isma Hadji, Hai X. Pham, Xiatian Zhu, Georgios Tzimiropoulos, and Brais Martinez. FAM diffusion: Frequency and attention modulation for high-resolution image generation with stable diffusion. *arXiv preprint arXiv:2411.18552*, 2024.

Zhihang Yuan, Hanling Zhang, Pu Lu, Xuefei Ning, Linfeng Zhang, Tianchen Zhao, Shengen Yan, Guohao Dai, et al. DiTFastAttn: Attention compression for diffusion transformer models. *arXiv preprint arXiv:2406.08552*, 2024.

Jinjin Zhang, Qiuyu Huang, Junjie Liu, Xiefan Guo, and Di Huang. Diffusion-4K: Ultra-high-resolution image synthesis with latent diffusion models. *arXiv preprint arXiv:2503.18352*, 2025a.

Peiyuan Zhang, Yongqi Chen, Runlong Su, Hangliang Ding, Ion Stoica, Zhengzhong Liu, and Hao Zhang. Fast video generation with sliding tile attention. *arXiv preprint arXiv:2502.04507*, 2025b.

Shen Zhang, Zhaowei Chen, Zhenyu Zhao, Yuhao Chen, Yao Tang, and Jiajun Liang. HiDiffusion: Unlocking higher-resolution creativity and efficiency in pretrained diffusion models. *arXiv preprint arXiv:2311.17528*, 2023.

Bingxuan Zhao, Qing Zhou, Yu Wang, Chuang Yang, and Qi Wang. σ: Sigmoid modulation for ultra high resolution diffusion. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 162993–163008. PMLR, 2026. URL https://proceedings.mlr.press/v306/zhao26cc.html.

Min Zhao, Bokai Yan, Xue Yang, Hongzhou Zhu, Jintao Zhang, Shilong Liu, Chongxuan Li, and Jun Zhu. UltraImage: Rethinking resolution extrapolation in image diffusion transformers. *arXiv preprint arXiv:2512.04504*, 2025. URL https://arxiv.org/abs/2512.04504.

Le Zhuo, Ruoyi Du, Han Xiao, Yangguang Li, Dongyang Liu, Rongjie Huang, Wenze Liu, Lirui Zhao, et al. Lumina-Next: Making Lumina-T2X stronger and faster with Next-DiT. *arXiv preprint arXiv:2406.18583*, 2024.

APPENDIX

This short edition of the appendix keeps the proofs, the analysis plots, the protocol in brief, and sample pages; the full version (62 pages) holds every table, interval, and prompt.

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A PROOFS

This appendix proves the statements that Sections 2 and 3 use. Lemma 1 is the split behind the definition of reach. Propositions 2 and 3 give the trained position range and the cost of the two views (Section 3.1). The proof of Proposition 1 gives the closed-form fit and the price of any coarser rule (Section 3.2). Remark 1 treats text-image offsets.

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into a near set $\mathcal{K}^{\text{near}}$ and a far set $\mathcal{K}^{\text{far}}$, with logits ℓ_j. Then $O^{\text{full}} = m\,O^{\text{near}} + (1-m)\,O^{\text{far}}$ with $m = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} / \sum_j e^{\ell_j}$ and $O^{\text{near}}, O^{\text{far}}$ the softmax outputs restricted to $\mathcal{K}^{\text{near}}$ and $\mathcal{K}^{\text{far}}$.*

Proof. Write $Z^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j}$, so that $Z^{\text{near}}O^{\text{near}} = \sum_{j\in\mathcal{K}^{\text{near}}} e^{\ell_j} v_j$, and likewise for the far set. Then $O^{\text{full}} = (Z^{\text{near}}O^{\text{near}} + Z^{\text{far}}O^{\text{far}})/(Z^{\text{near}} + Z^{\text{far}})$, and $Z^{\text{near}}/(Z^{\text{near}} + Z^{\text{far}}) = m$. □

The lemma motivates the two-branch form; SCA's key sets are not a partition. Block centres can fall inside the window, and text keys enter both branches. A bridge token inside the window appears twice in the local softmax, which doubles its weight there.

Proposition 2 (Trained position range)**.** *Assume exact integer positions and $t \le n_0 \le \min(H, W)$. Every query-key offset of the global operator, and every offset between an image query and a key of its native window $\mathcal{W}_{n_0}(i)$ (bridge keys and wider windows excluded), is at most $n_0 - 1$ per axis. Each global-operator query sees the native $T + n_0^2$ keys.*

Proof of Proposition 2. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \ge w$ and a tile $[t\tau,\, t\tau + t)$ with $t \le w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \le t\tau$ because $\lfloor t/2 \rfloor \le \lfloor w/2 \rfloor$, and $a' + w \ge t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \ge \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \ge a' + w \ge t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \le t\tau$ and $H \ge t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \le n_0$ differ by at most $n_0 - 1$. The proposition makes no claim about the bridge keys, which keep their true positions. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and in AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proposition 3 (Cost)**.** *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N + T)(T + N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T + R + w^2)$ with $R = |\mathcal{B}|$. For fixed n_0, w, R, and T the total is $O\big(N(T + R + w^2 + N_0)\big)$, linear in N.*

Proof of Proposition 3. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys, R bridge keys, and w^2 window keys. The sum is at most $(2N + T)(T + R + w^2 + N_0)$, which is $O(N(T + R + w^2 + N_0))$ for fixed n_0, w, R, and T. Value aggregation has the same pair counts. In the opening phase the local term is absent.

The FlexAttention kernel visits whole blocks and masks the extra pairs. A query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 1. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\text{loc}} = O^{\text{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\text{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \leq 0$, so $A\pi^\star - B \geq 0$ and $\pi' - \pi^\star \geq 0$. If $\pi^\star = 1$ then $B \geq A$, so $A\pi^\star - B \leq 0$ and $\pi' - \pi^\star \leq 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell is a feasible π' in every cell. Examples are a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere. Its total error is therefore at least the per-head table's plus $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$. A table interpolated in σ between the mean sampled noise levels of occupied bins is not constant within a bin. The statement applies to the bin-constant table used to score rules in Section 4.3. □

B REACH EMULATION: DETAILS AND STATISTICS

This appendix expands Section 2, Figure 2, and Section 3.2 with Figure 4. Appendix B.1 gives the noise bins. Appendix B.2 gives the attention mass behind the observation, Appendix B.3 the fitted reach tables, and Appendix B.4 the output error of every mixing rule.

B.1 NOISE BINS

The latent at noise level σ is $x_\sigma = (1 - \sigma)x_0 + \sigma\epsilon$. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105. The lowest of the 14 bins is therefore empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

B.2 THE HEAD SETS THE REACH

Figure B.1 ties each head's measured share to its fitted weight.

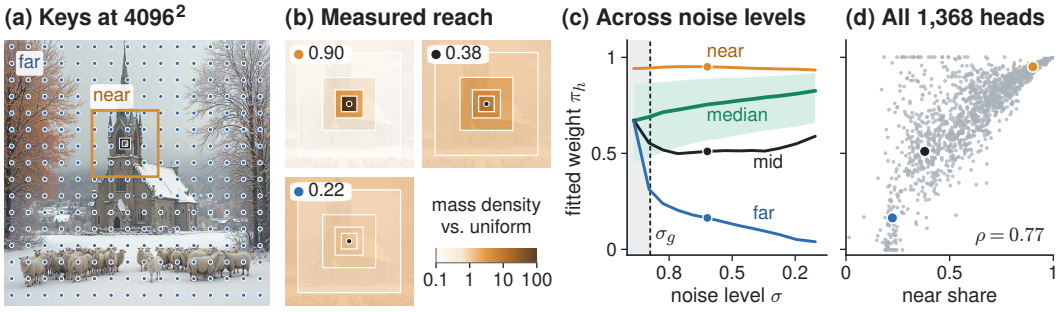


Figure B.1: **Measured reach tracks the fitted weight π_h.** (a) Near window (amber) and every fourth block-centre key per axis of the far view (blue) on a 4096^2 output. (b) Full-attention mass density of three heads at 1024^2 and $\sigma = 0.62$ (log scale, clipped at 0.1 and 100 times uniform), with each head's near share, its share of image attention within 16 tokens. (c) Their π_h with the median and interquartile range of all 1368 heads; σ falls left to right; dashed: the gate σ_g; grey band: the far-only opening. (d) π_h against the near share, Pearson $\rho = 0.77$.

Attention mass. For 256 seeded image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass. Image keys within distance

16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$. Keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step, even at native resolution. Figure B.1b and d use the per-head shares in the noise bin at $\sigma = 0.62$.

B.3 The reach table

The reach table holds one weight per block, head, noise bin, and ratio. Figure B.2 shows the ratio-4 table that sampling reads and how well a table transfers to another ratio or resolution. Figure B.3 plots every head's weight against noise level with the split-half agreement, and Figure B.4 shows how widely the heads of one block differ.

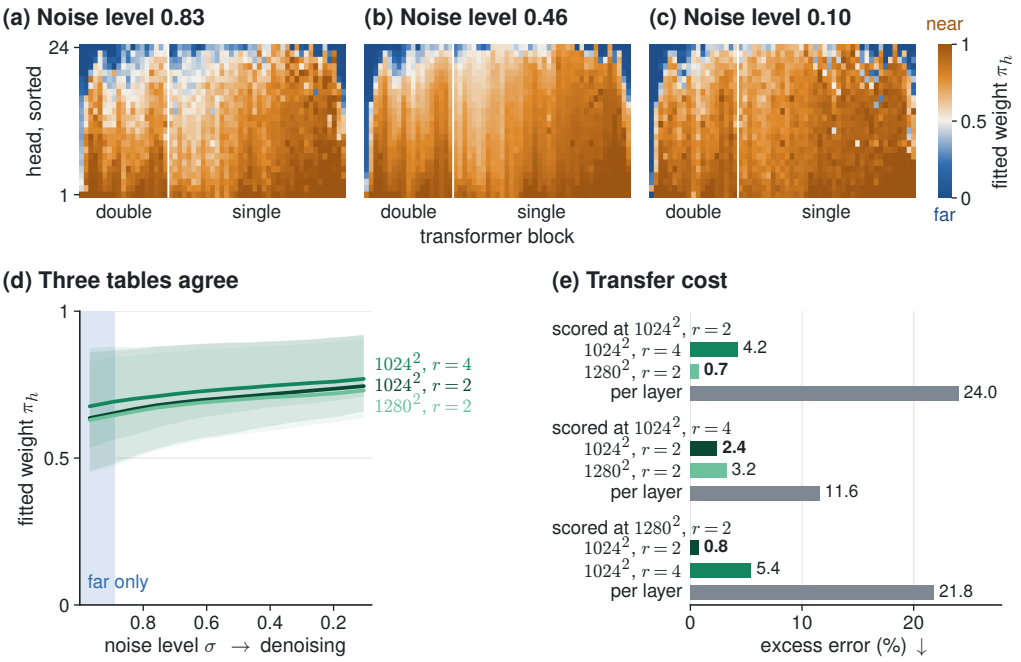


Figure B.2: **Heads differ widely, yet a reach table transfers across ratio and resolution.** (a to c) $\pi_h(\sigma)$ of the ratio-4 reach table at $\sigma = 0.83$, 0.46, and 0.10. (d) Mean and interquartile range across heads of three tables; shaded: far-only opening. (e) Excess error over each setting's own table: at most 5.4% transferred, 12 to 24% per layer.

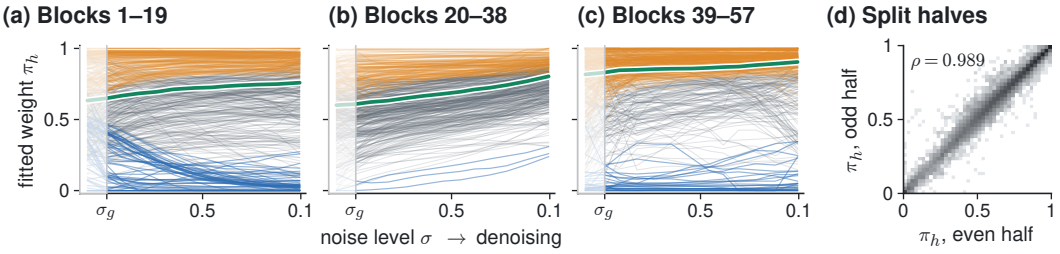


Figure B.3: **Heads use different mixes, and the mix repeats on new prompts.** (a to c) $\pi_h(\sigma)$ of all 1368 heads at ratio 2: blocks 1 to 19 (double-stream), 20 to 38, and 39 to 57 (single-stream); amber: mean above 0.8, blue: below 0.2, green: median; $\sigma > \sigma_g$ shaded. (d) Tables fitted on disjoint prompt halves agree (Pearson $\rho = 0.989$).

B.4 Output error of each mixing rule

Figure B.5 compares the relative output error of every mixing rule at both ratios, and Figure B.6 resolves it by noise level; Section 4.3 quotes these errors. A relative error is $\sum J / \sum \|O^{\text{full}}\|^2$, pooled over blocks, heads, and steps, on 32 prompts at 1024^2 and 16 at 1280^2.

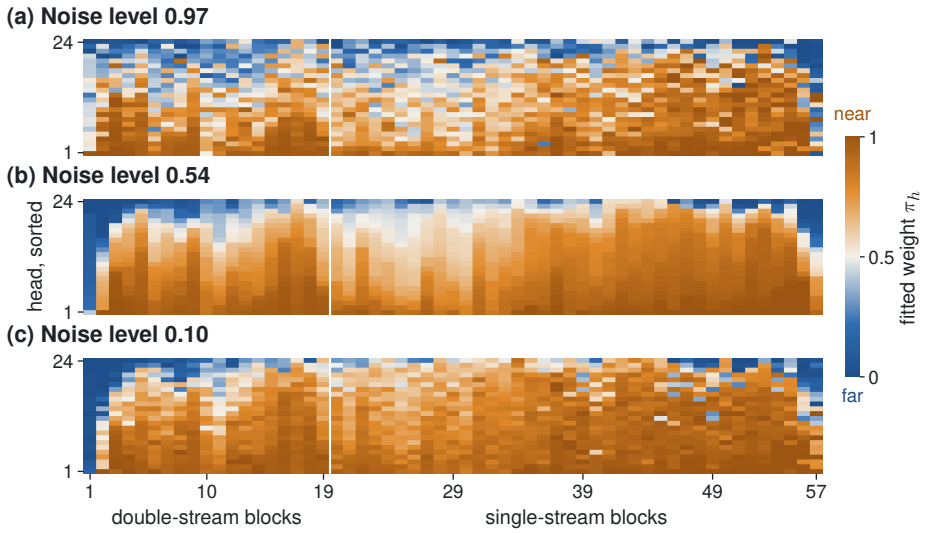


Figure B.4: **Heads in the same block differ widely at every noise level.** $\pi_h(\sigma)$ of the ratio-2 table at $\sigma = 0.97$ (a), 0.54 (b), and 0.10 (c): 57 blocks by 24 heads, 32 prompts, heads sorted per block by (b). The median within-block range is 0.66 to 0.84. Figure B.2 shows the ratio-4 table that sampling reads.

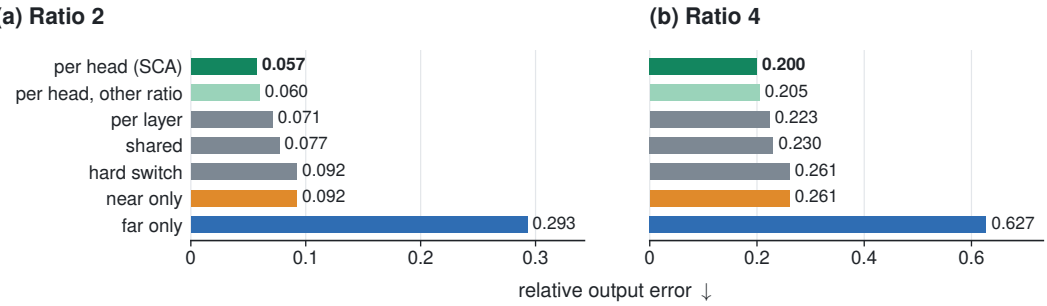


Figure B.5: **Per-head weights explain full attention best, and they transfer across ratios.** Relative output error of each mixing rule at emulated ratios 2 (a) and 4 (b), 32 prompts. The other-ratio rule applies the table fitted at the other ratio. The best hard switch ($\sigma^\star = 1$) equals the near view (local branch) alone. In-sample and held-out errors agree within 0.0003.



Figure B.6: **In output energy, no noise level prefers the far view.** (a) Fraction of heads per weight π and noise bin, with the per-head mean, the best shared $\pi(\sigma)$, and AReA's switch; ratio 2, 32 prompts; σ falls left to right, from pure noise at $\sigma = 1$. (b) Relative error per noise bin: the near view alone beats the far view alone everywhere, and the per-head mix beats both.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure B.6). The median within-block range of the per-head weights is 0.66 to 0.84 at every noise level (Figure B.4). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at ratio 2 and from 0.261 to 0.376 at ratio 4. In the noisiest bin the global branch alone has relative error 0.27 at ratio 2, the local branch alone 0.14, and the per-head mix 0.09. Generation still needs the opening (Section 4.3), because output error cannot see layout.

C Implementation

This appendix expands Section 3, Figure 3, and Algorithms 1 and 2. Figure C.1 draws the operators at component level (Appendix C.1), and the full version gives the pseudocode, the kernel, the cost, and every hyperparameter.

C.1 The operators in detail

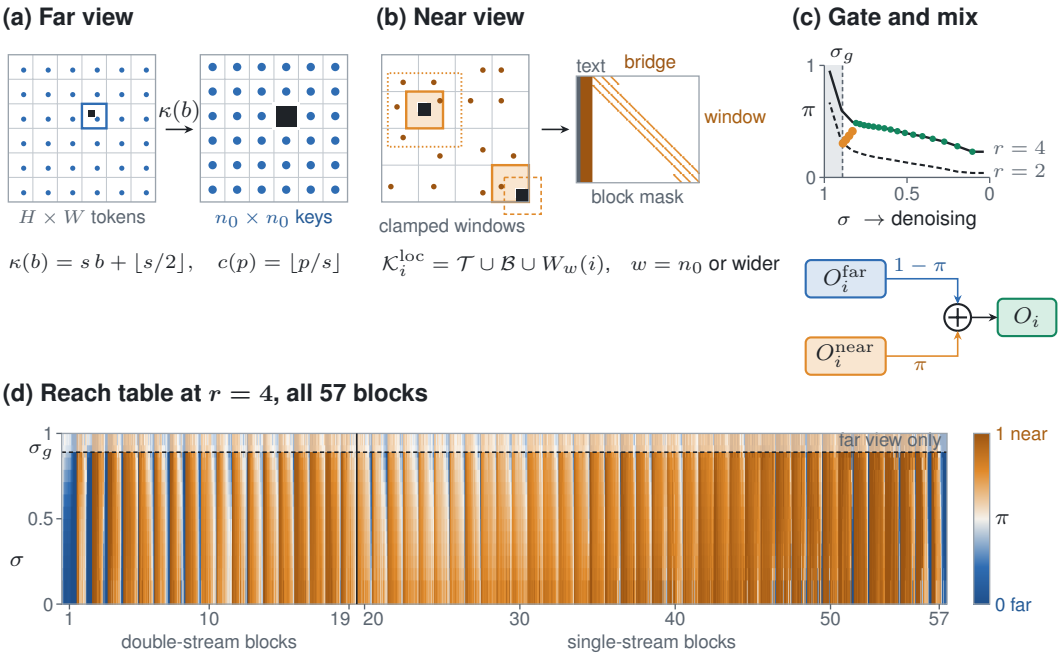


Figure C.1: **Both views keep the native geometry, and one weight per head and step mixes them.** The operators of Figure 3 at component level on a toy grid: $H = W = 24$ tokens, $s = 4$, $n_0 = 6$, $t = 2$. (a) The far view keys each block by its centre token (Equation 4). (b) A query tile's near view: text keys, bridge tokens (dots), and a border-clamped window, with its block mask (Equations 2 and 3). (c) Head 24 of block 12 at $r = 2$ (dashed) and $r = 4$ (solid); dots: the weight applied at each post-gate step at 4096^2 (amber: wide windows; σ falls left to right), and the mix (Equation 6). (d) The $r = 4$ table, 57×24 heads by 14 noise bins.

Window placement. Image queries form tiles of $t \times t$ tokens; query i lies in tile $\tau = \lfloor p_i/t \rfloor$. The window $\mathcal{W}_w(i)$ of tile τ has side w, at most the grid side, and starts at $a(\tau) = \mathrm{clip}\left(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor,\ 0,\ H - w\right)$ per axis. It is the $w \times w$ block centred on the tile and clamped to the grid, and it contains the whole tile (proof of Proposition 2).

D Evaluation protocol in brief

Section 4 gives the data and the metrics; this appendix adds how each baseline was run and how intervals and times are computed. The full version defines every metric and gives each baseline's code revision.

Baselines. AReA runs its released code with its FLUX.1 defaults on FLUX.1-dev, with the seeds shared by all methods; rerun on FLUX.1-Krea-dev, its default checkpoint, it reproduces its published scores. DyPE and SEGA run through our frozen baseline runner on the same sampler. UltraImage runs its two-stage defaults: a 1024^2 and a 4096^2 stage of 30 steps each, at guidance 3.5 and then 7. HRDiT runs its released progressive protocol in fp16: 30 steps at 1024^2, 17 at 2048^2, and 10 at 4096^2, at guidance 3.5, 4.5, and 6.0. Progressive SCA runs HRDiT's schedule in bf16 with SCA attention in the two high-resolution stages, which read the table at ratios 2 and 4. The attention, the precision, and the runner change together, so the comparison with HRDiT does not isolate the attention. SigMa runs its released code with a profile matched to FLUX.1-dev. FLUX.2-dev uses its own reach table, 28 steps at guidance 4.0, and a shift fixed to its native sequence length.

Intervals and multiple comparisons. Intervals are 95% paired percentile bootstraps over prompt ids, with 1000 or 2000 draws as each caption states; for FID, KID, and their patch versions the draws also resample the 168 distinct reference images. Table E.2 gives the headline comparisons before and after a Holm correction over the ten per-image metrics. The intervals measure prompt and reference sampling, not variation over training runs.

Timing. Clean latency is the median of three timed images at batch 1 on an idle GH200 after one excluded warm-up image. It covers text encoding, every denoising step, and VAE decoding, but not model loading. Progressive SCA, HRDiT, and AReA were timed back to back in one run (run T3). Campaign means from the shared evaluation nodes are marked and never ranked against clean times.

E Results in detail

This appendix adds to Section 4. Table E.1 expands Table 1 to every metric, Table E.2 gives the Holm-adjusted calls, and Figures E.1 to E.4 show crop detail, the baseline differences, latency, and quality against time. Every table follows the conventions of Appendix D.

E.1 Every metric of Table 1

Table E.1 expands Table 1 to all 14 metrics and adds the two non-square sizes and 6144^2. Each row is marked against AReA, and the full version gives the intervals.

E.2 Multiple comparisons

Table E.2 gives, for each headline pair, the difference on every per-image metric and its p-value before and after a Holm correction over the ten per-image metrics. Against AReA, progressive SCA stays significantly better on ten of the ten per-image metrics and single-pass SCA on ten of ten. Against HRDiT, two differences survive, HPSv2 and ARNIQA, both in favour of progressive SCA; MUSIQ (in its favour) and CLIP-IQA (against it) do not.

Where the four methods draw the same content, progressive SCA's crop detail is level with HRDiT's (Figure E.1).

E.3 Every baseline against single-pass SCA

Figure E.2 plots four metrics of single-pass SCA minus each baseline on the same 195 prompts. Every shared-sampler baseline trails single-pass SCA on all ten per-image metrics, and HRDiT and UltraImage, which run their own multi-stage recipes, lead on several preference metrics (full version).

E.4 Latency

Figure E.3 expands the latency of Figure 7 and Section 4.2, and Figure E.4 plots ImageReward against time. All times follow Appendix D.

Table E.1: **Both SCA modes beat AReA on all ten per-image metrics at 4096^2; the non-square sizes are mixed.** Expands Table 1 to every metric and adds the non-square sizes and 6144^2. FLUX.1-dev, 195 prompts (100 at 6144^2, where the row is SCA without the schedule); HPS, CLIP-IQA, and ARNIQA $\times 100$; AReA runs its released code. Blue / orange: the 95% paired percentile-bootstrap interval against AReA excludes zero in favour of / against the row. Progressive SCA is paired over 2000 prompt draws, and for FID and FID_p over 1000 draws that also resample reference-image clusters. Full attention's FID and FID_p are unmarked. **Bold** / underline: best / second best per size (bold only in the two-row blocks); shaded: ours. Time: batch-1 median of three on an idle GH200 (HRDiT and progressive SCA: run T3, Appendix D); ‡campaign mean on shared nodes; °UltraImage's two-stage defaults. Key result: at 4096^2 progressive SCA is the fastest method (49.2 s) and best on CLIP, both HPS scores, and ARNIQA; single-pass SCA is best on patch FID and both MUSIQ scores.

	Time	Preference ↑						Image quality ↑				Set level ↓	
Method	s ↓	IR	Aesth.	CLIP	Pick	HPSv2	HPSv2.1	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA	FID	FID_p
4096 × 4096, *scale factors (4, 4)*													
Full attention	186.5	−2.18	4.50	15.37	17.94	20.75	7.60	22.9	33.1	49.0	54.4	312.4	252.2
DyPE	195.9‡	−0.10	5.72	24.92	20.51	25.53	20.55	36.3	50.3	59.6	60.2	205.5	171.9
SEGA	190.4‡	0.56	6.02	26.95	21.64	27.13	24.34	37.3	50.3	66.7	64.8	164.9	125.8
UltraImage°	308.5‡	1.28	6.50	28.84	23.33	29.96	30.77	33.6	41.7	72.8	72.3	142.0	109.5
HRDiT	107.0	**1.34**	**6.53**	29.12	**23.46**	29.88	30.92	40.3	53.5	**75.8**	71.6	**138.4**	103.0
AReA	202.4	0.92	6.27	27.69	22.36	28.62	27.16	39.4	50.4	67.7	67.4	145.8	102.9
Single-pass SCA	77.1	1.25	6.42	29.14	23.12	29.77	29.73	**45.0**	**60.1**	72.3	70.7	140.2	**98.4**
Progressive SCA	**49.2**	1.32	6.51	**29.39**	23.41	**30.26**	**31.02**	41.6	54.1	74.9	**74.5**	141.9	107.7
2048 × 4096, *scale factors (2, 4)*													
AReA	112.4‡	**1.13**	**6.41**	**28.59**	**22.66**	28.36	**27.64**	51.9	61.6	**74.9**	**73.0**	144.6	**111.7**
Single-pass SCA	**41.1**‡	1.11	6.17	28.09	22.61	**28.57**	27.49	**56.2**	**65.3**	72.5	72.6	**142.5**	114.9
2048 × 6144, *scale factors (2, 6)*													
AReA	150.5‡	0.39	**6.14**	26.52	21.51	26.41	23.36	42.8	53.0	67.5	69.3	163.0	**105.7**
Single-pass SCA	**60.5**‡	**0.64**	5.88	**26.78**	**21.66**	**26.59**	**23.60**	**47.7**	**59.4**	**67.8**	**69.5**	**142.8**	109.7
6144 × 6144, *scale factors (6, 6), 100 prompts*													
AReA	507.9‡	−0.42	5.80	23.84	20.30	24.95	18.78	28.3	44.5	53.8	47.8	216.7	188.6
SCA w/o schedule	**164.7**‡	**0.84**	**6.24**	**27.12**	**21.89**	**27.48**	**25.16**	**37.5**	**48.3**	**60.1**	**62.6**	**169.9**	**122.4**

Table E.2: **Every win over AReA survives Holm correction; against HRDiT, two differences of four do.** FLUX.1-dev at 4096^2. Paired differences, method minus reference, with two-sided bootstrap p-values from the same 2000 prompt draws (seed 0) that give every interval of the paper, before and after Holm's step-down adjustment over the ten per-image metrics of each comparison. Blue / orange: significantly better / worse after adjustment at 5%. $p \leq 0.001$ is the smallest value 2000 draws resolve. The full version adds the record of every paired comparison of Tables 1 to 3, with a sign-flip check. Key result: progressive and single-pass SCA keep all ten wins over AReA; against HRDiT, HPSv2 and ARNIQA remain better and MUSIQ and CLIP-IQA no longer differ.

	Progressive − AReA			Progressive − HRDiT			Single-pass − AReA		
Metric	Δ	p	p_Holm	Δ	p	p_Holm	Δ	p	p_Holm
Preference									
ImageReward	+0.399	≤0.001	0.010	−0.022	0.453	1.000	+0.327	≤0.001	0.010
Aesthetic	+0.23	≤0.001	0.010	−0.03	0.230	1.000	+0.15	≤0.001	0.010
CLIP score	+1.70	≤0.001	0.010	+0.27	0.056	0.336	+1.45	≤0.001	0.010
PickScore	+1.05	≤0.001	0.010	−0.05	0.249	1.000	+0.76	≤0.001	0.010
HPSv2 (×100)	+1.64	≤0.001	0.010	+0.38	≤0.001	0.010	+1.15	≤0.001	0.010
HPSv2.1 (×100)	+3.86	≤0.001	0.010	+0.10	0.406	1.000	+2.56	≤0.001	0.010
Image quality									
MUSIQ	+2.3	0.009	0.010	+1.3	0.030	0.240	+5.6	≤0.001	0.010
MUSIQ_c	+3.7	≤0.001	0.010	+0.6	0.330	1.000	+9.7	≤0.001	0.010
CLIP-IQA (×100)	+7.2	≤0.001	0.010	−0.9	0.035	0.245	+4.6	≤0.001	0.010
ARNIQA (×100)	+7.1	≤0.001	0.010	+2.9	≤0.001	0.010	+3.4	≤0.001	0.010

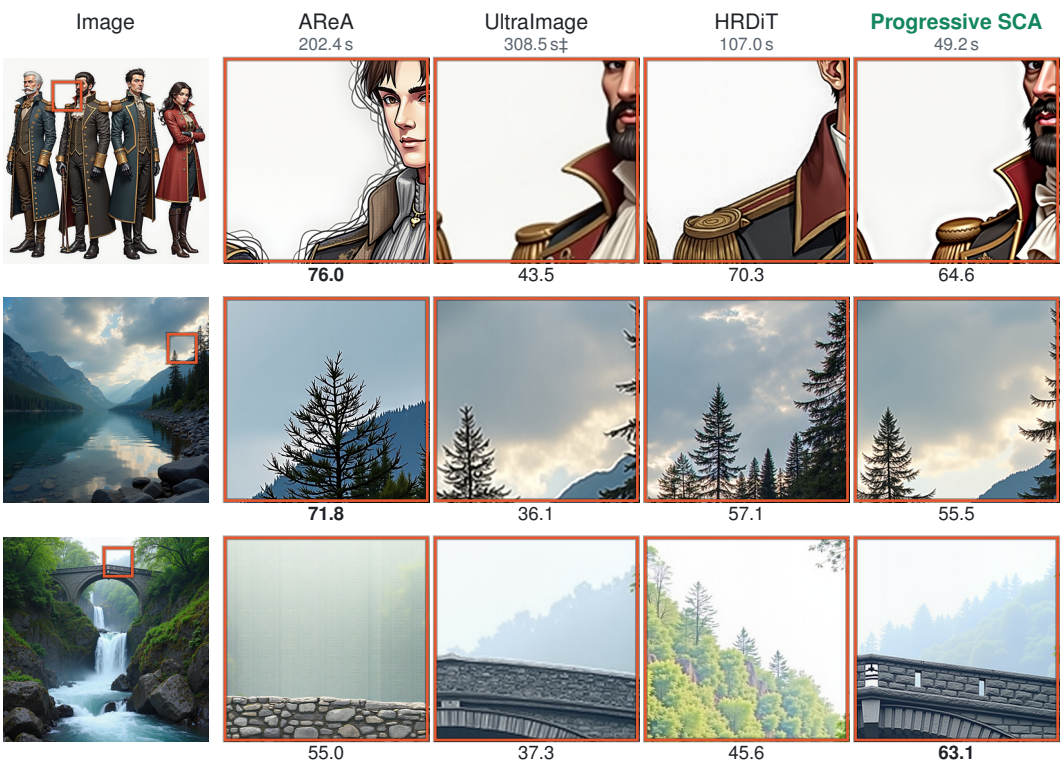


Figure E.1: **Where the four methods draw the same content, progressive SCA's crop detail is level with HRDiT's.** Per category, the evaluation prompt and 512×512 window on which the four methods agree most in content among high-detail windows, seed 0; AReA's and HRDiT's own layouts still differ in some rows. Numbers: MUSIQ of each crop, best bold; times: clean seconds per 4096^2 image (‡ campaign mean). On these three crops MUSIQ is lowest for UltraImage in every row and highest for AReA in two. Over all 195 prompts, progressive SCA's MUSIQ over four 1024^2 crops is indistinguishable from HRDiT's ($+0.6$ $[-0.7, 1.8]$) and exceeds AReA's by $+3.7$ $[2.0, 5.4]$ and UltraImage's by $+12.4$ $[11.5, 13.2]$ (full version).

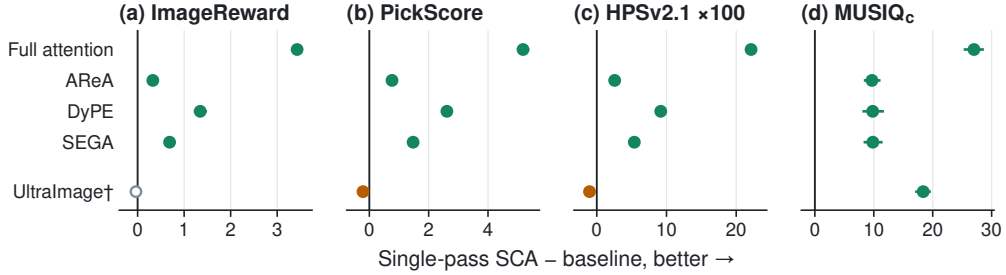


Figure E.2: **Single-pass SCA beats every shared-sampler baseline on all four metrics; against UltraImage the result is mixed.** Single-pass SCA minus each baseline on the same 195 prompts at 4096^2, seed 0; whiskers are 95% paired intervals over prompts (2000 bootstrap draws), not adjusted for multiplicity. Filled: the interval excludes zero (green: SCA better, orange: worse); open grey: it includes zero. Against UltraImage, SCA is better on MUSIQ_c and worse on PickScore and HPSv2.1; †UltraImage runs its own two-stage recipe, so this is not an equal-cost comparison. Table E.1 gives all metrics.

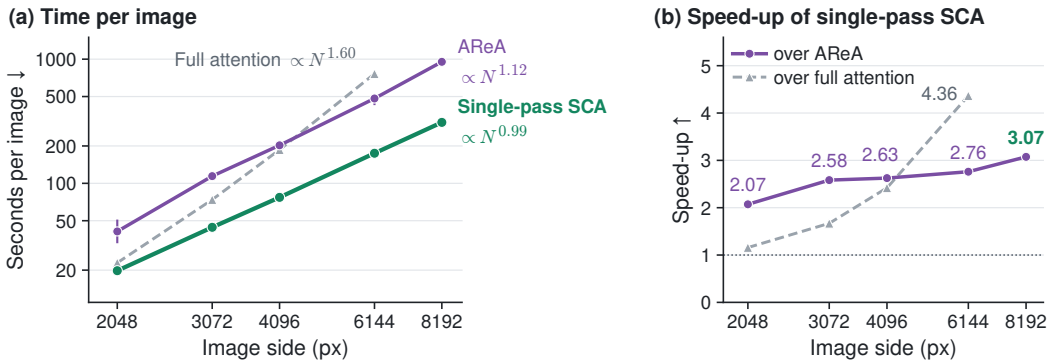


Figure E.3: **Single-pass SCA is the fastest method at every size; its lead over AReA grows to 3.07× at 8192^2.** (a) Median seconds per image of three timed images after one excluded warm-up (batch 1, one GH200, 28 steps including text encoding and VAE decoding), whiskers min to max; labels give b of seconds $\propto N^b$ (N image tokens). (b) Each baseline's median time over single-pass SCA's; dotted: parity. Full attention was not run at 8192^2.

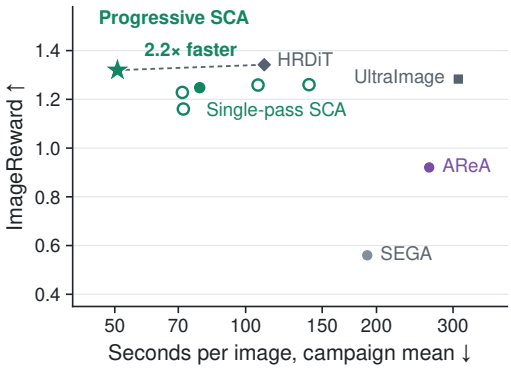


Figure E.4: **At 46% of HRDiT's time, progressive SCA is indistinguishable from HRDiT on ImageReward (difference -0.02 $[-0.09, 0.03]$).** ImageReward against campaign seconds per image on shared nodes, $n = 195$ prompts at 4096^2. Star: progressive SCA; filled circle: single-pass SCA; open circles: SCA without the schedule at 28, 42, and 56 steps and at guidance 7; the dashed guide carries the ratio of campaign times. Full attention, DyPE, and SigMa fall below the range (full version).

F Limitations and broader impact

Limitations. Only the mixing weight is fitted; the gate, the window sizes, the tile, the bridge, the emulated ratios, and the noise bins are set by hand, and the window sizes were chosen on the 46-prompt subset, part of the evaluation set. Progressive SCA has a higher patch FID than HRDiT and single-pass SCA, it depends on HRDiT's sampler, and the 6144^2, 8192^2, FLUX.2-dev, and sensitivity results use SCA without the schedule. Reach emulation omits the bridge and sees 16 times fewer keys per view than the target. No-reference quality metrics reward sharpness even where the layout fails, so we report every metric and sample pages; the full version shows failure cases.

Broader impact. SCA shortens generation at 4096^2 relative to full attention under our measured protocol, which helps applications such as print, design, and scientific visualization. Energy use remains unmeasured. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

G Sample pages

This appendix adds full pages of samples to Figures 1 and 6; each caption states how its prompts were chosen. Figure G.1 sets both modes of SCA beside AReA, UltraImage, and HRDiT, and Figure G.2 shows eight methods on two more prompts. Figure G.3 shows single-pass SCA at three aspect ratios, Figure G.4 sets it beside AReA, and Figure G.5 shows the ablation variants of Table 3.

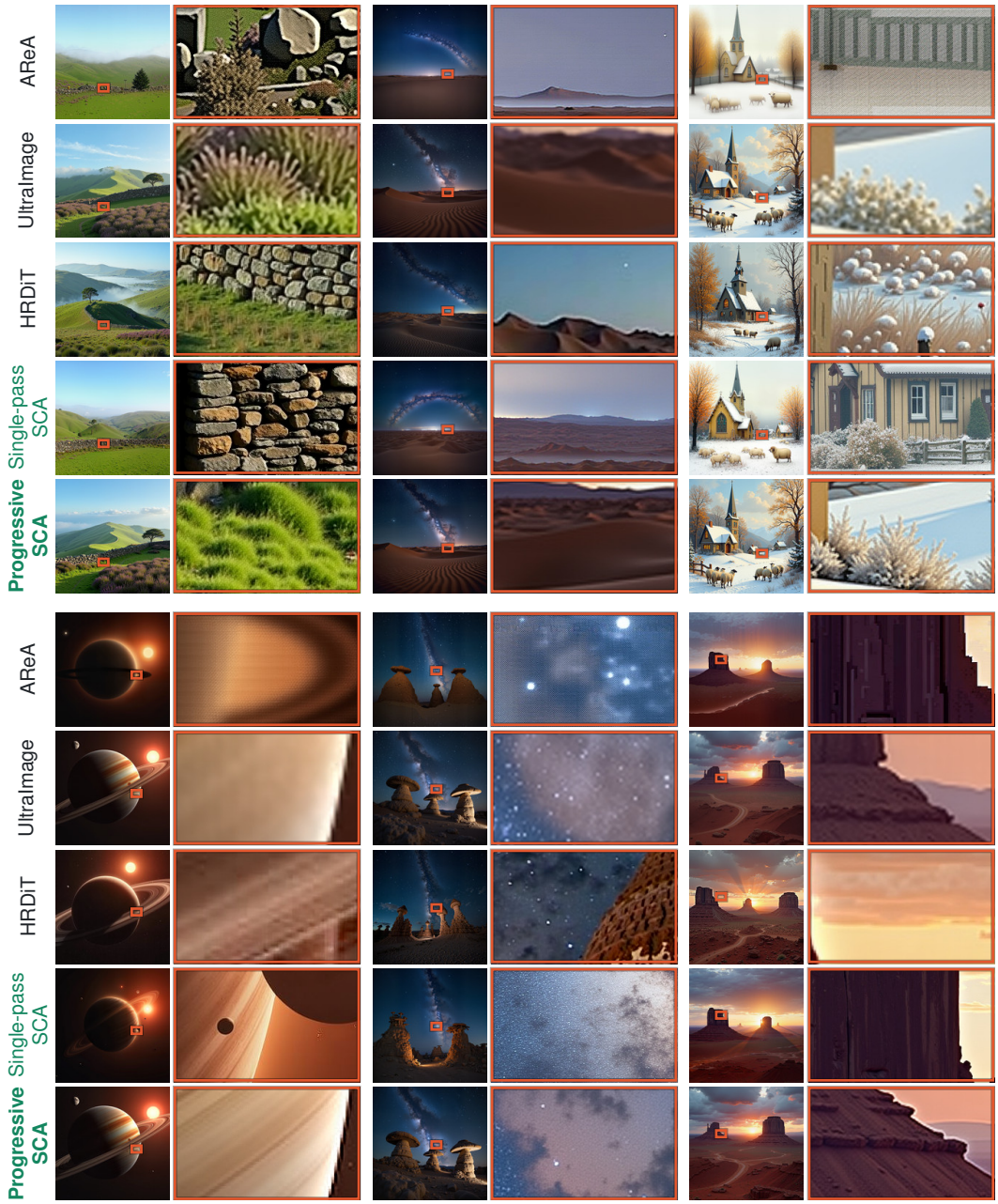


Figure G.1: **Both modes of SCA beside AReA, UltraImage, and HRDiT at 4096^2.** Six evaluation prompts that name no person (keyword rule) outside Figure 6, chosen by a fixed seed (seed 0); seed 0 for every image, 4096^2. Each crop shows native pixels of the window that maximises the smallest edge energy among the five methods. Over all 195 prompts, progressive minus single-pass SCA is -6.0 $[-7.3, -4.6]$ on MUSIQ over four 1024^2 crops and $+0.07$ $[0.01, 0.13]$ on ImageReward (full version).

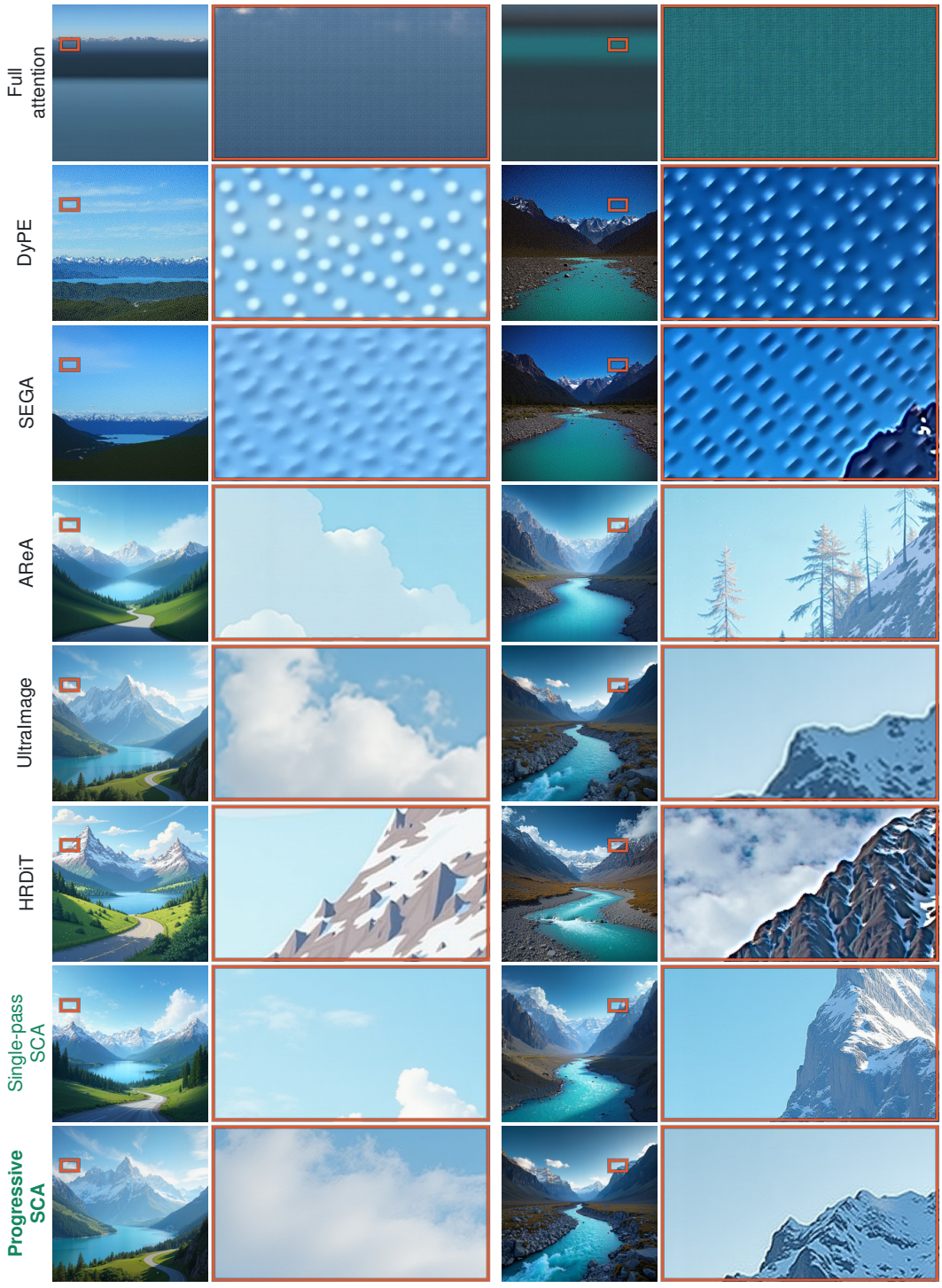


Figure G.2: **Full attention loses the scene and DyPE and SEGA overlay a lattice; the other five keep it.** 4096^2, evaluation prompts that name no person (keyword rule) not shown in an earlier sample figure, chosen by a fixed seed (seed 8); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among the eight methods. Over all 195 prompts, full attention, DyPE, and SEGA each trail progressive SCA on all ten per-image metrics (Table 1).

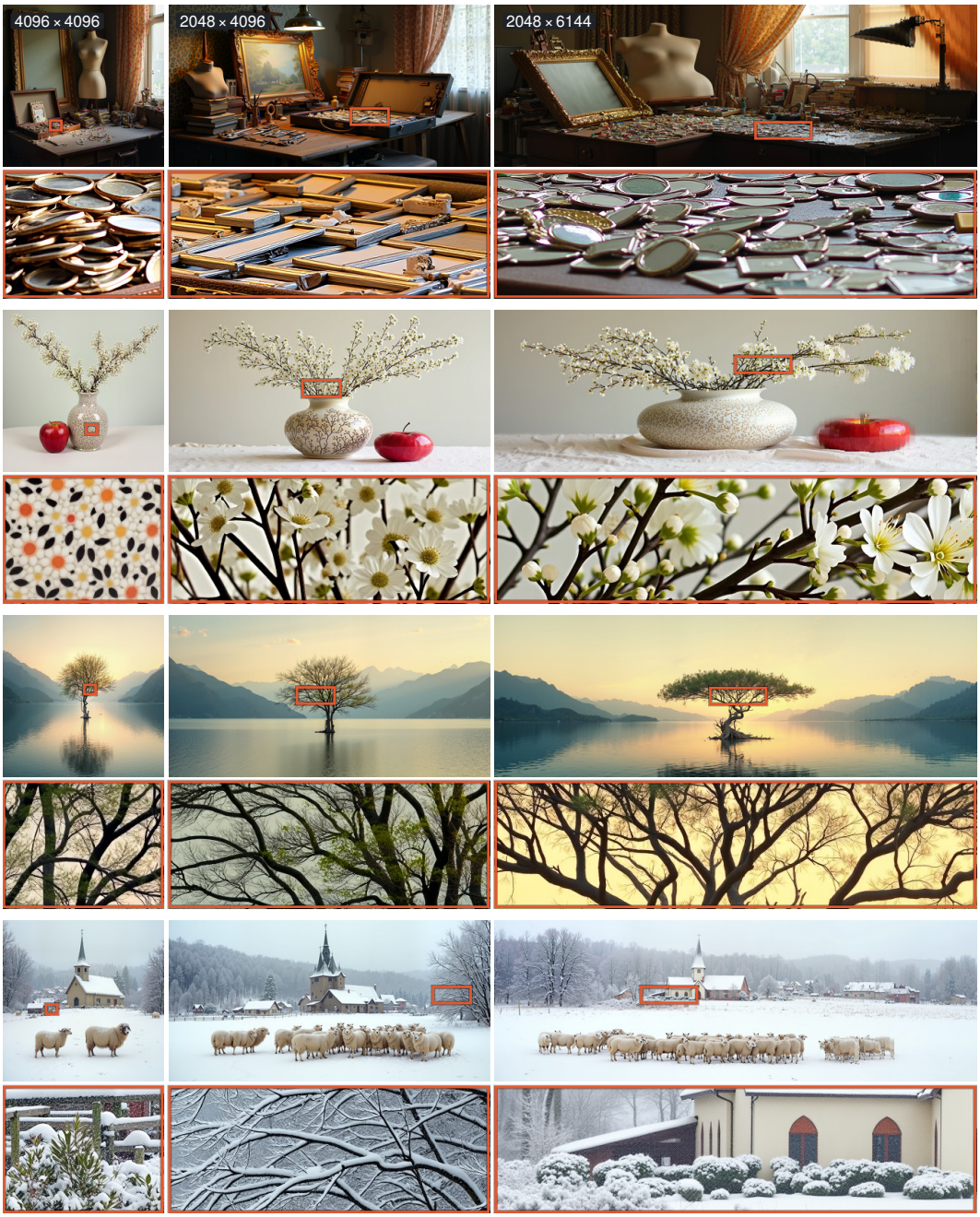


Figure G.3: **Single-pass SCA at three aspect ratios with one reach table.** Single-pass SCA on FLUX.1-dev at 4096^2, 2048×4096 and 2048×6144 for four evaluation prompts that name no person (keyword rule) and no quoted text, chosen by a fixed seed (seed 3); seed 0 for every image. Each crop shows native pixels of the red box above it, at the highest edge energy of that image. Against AReA, single-pass SCA wins seven of 15 metrics at 2048×6144, and wins three and loses three at 2048×4096 (full version).

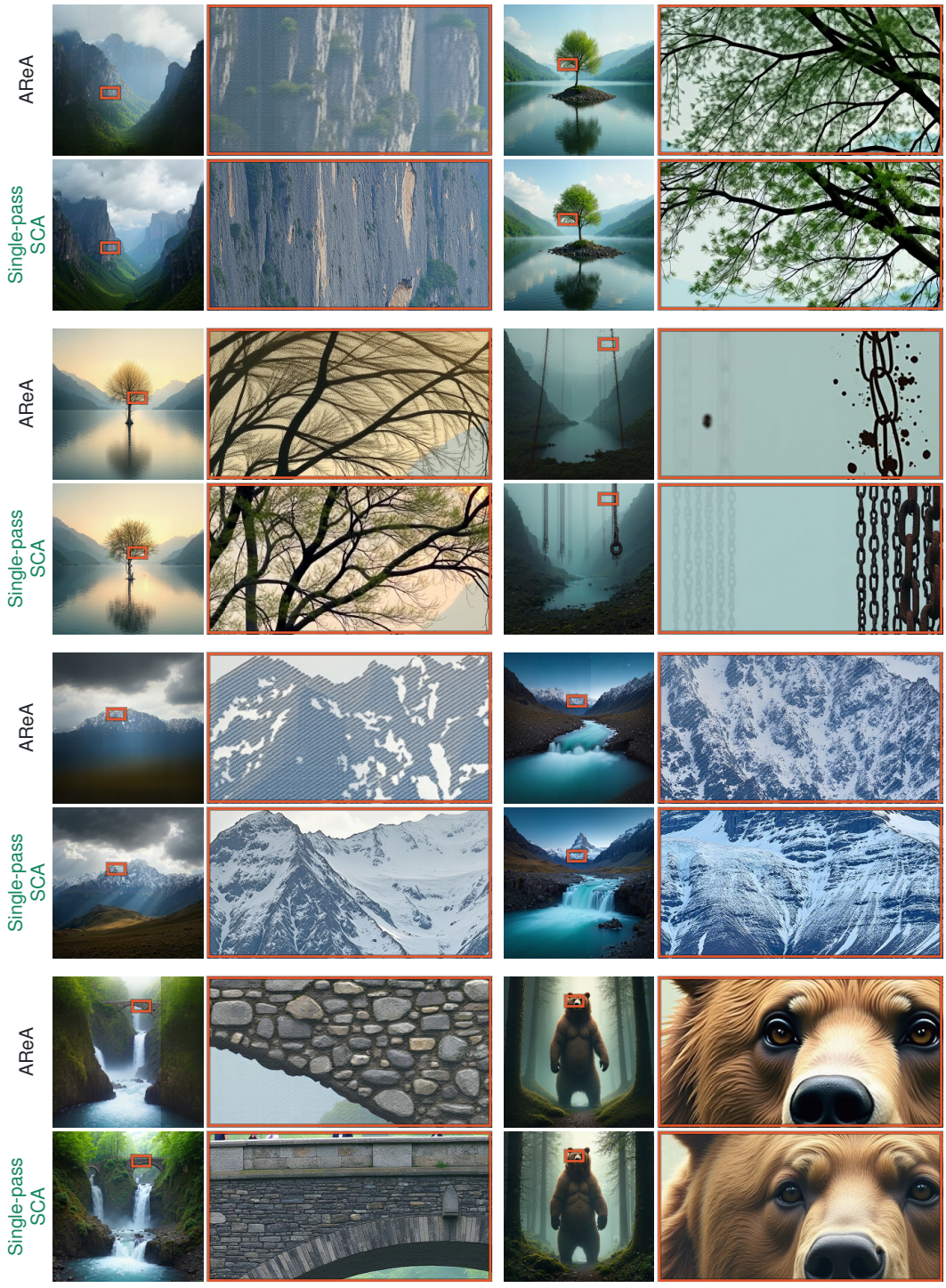


Figure G.4: **Single-pass SCA beside AReA at 4096^2.** Evaluation prompts that name no person (keyword rule) not shown in an earlier sample figure, chosen by a fixed seed (seed 0); seed 0 for every image. Each crop shows native pixels of the window that maximises the smallest edge energy among both methods. Over all 195 prompts, single-pass SCA is significantly better than AReA on all ten per-image metrics, both MUSIQ scores among them (Table E.1).

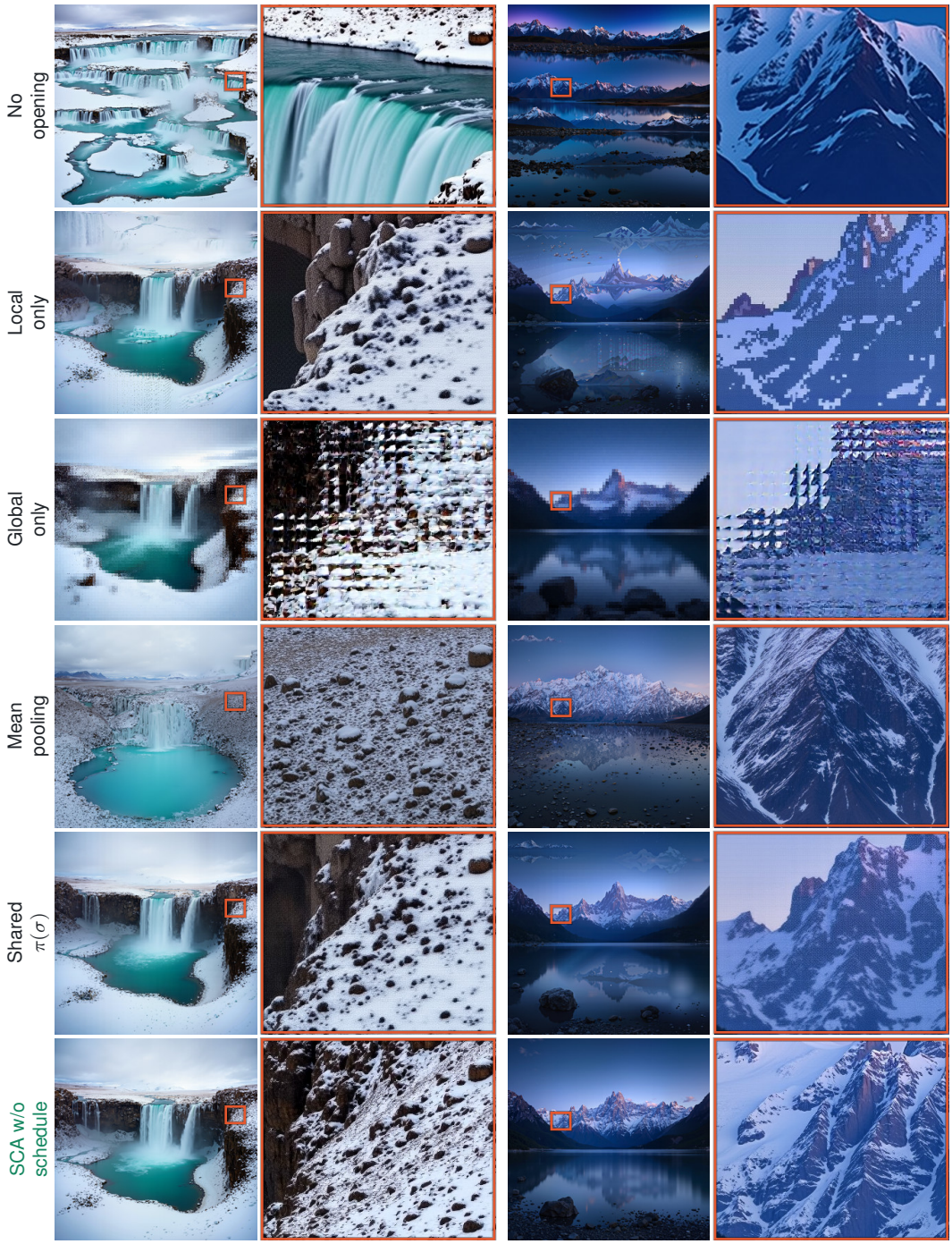


Figure G.5: **Without the opening phase, subjects shrink to a small scale.** Two of the 46 ablation prompts that name no person (keyword rule), chosen by a fixed seed (seed 46); 4096^2, seed 0. Each row changes one part of SCA without the schedule (Table 3): local only flattens texture into blocks, global only leaves a grid, and mean pooling moves the layout. Each crop shows native pixels of the window that maximises the smallest edge energy among the six variants. Over the 46 ablation prompts, dropping the opening phase changes ImageReward by -1.33 $[-1.60, -1.06]$ (Table 3).